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Resilient MR-based human-swarm interaction for UAV search and rescue in risk-conflict scenarios

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ABSTRACT

With the growing use of unmanned aerial vehicles (UAVs) in search and rescue (SAR) missions, improving the efficiency and safety of UAV swarm operations has become increasingly important. Operators often face complex and unstructured risk-conflict situations during SAR missions, which challenge their situational awareness and psychological resilience. This study proposes a resilient mixed reality (MR) human–swarm interaction (HSI) system designed to provide more intuitive and immersive interaction in high-risk environments, thereby enhancing operators' situational awareness and psychological resilience. Building on the team situational awareness model, this research incorporates resilience design principles. First, it analyzes the sources of risk and conflict in wilderness SAR scenarios and identifies the cognitive characteristics of human–swarm teams. Based on a resilience interaction model, an MR HSI system was developed to provide enhanced perceptual feedback, compensating for operators' cognitive limitations and improving both psychological resilience and system-level risk tolerance. Finally, an experiment was conducted using a prototype MR HSI system in representative risk-conflict scenarios. The results show that the proposed approach significantly improves operators' situational awareness and effectively supports psychological resilience in high-risk SAR tasks.

1. Introduction

Autonomous and remotely operated robots are playing an increasingly important role in civilian applications. Wilderness search and rescue (SAR) missions, in particular, have benefited significantly from the integration of Human–Robot Teams (HRTs). Robotic teammates allow responders to access uncertain and hazardous environments while extending human operational capabilities. Compared with single-robot systems, multi-robot teams provide situational awareness from multiple vantage points (Sato et al., 2004), enable the deployment of distributed sensors with reduced payload requirements (Parker, 2007), and improve mission fault tolerance (Liu and Nejat, 2016) .

Unmanned aerial vehicles (UAVs) have become a key aerial resource in SAR missions due to their high maneuverability, rapid deployment, and operational flexibility (Pajares, 2015). Their use significantly reduces response time and increases mission coverage. To fully exploit the advantages of UAV swarms, effective Human–Swarm Interaction (HSI) is required. As a subfield of Human–Computer Interaction (HCI), HSI focuses on how humans guide, coordinate, and supervise robot swarms while preserving swarm autonomy to achieve shared mission goals (Benabbou et al., 2023; Kolling et al., 2015).

Despite the advantages of UAV swarms in SAR missions, operating them in wilderness environments remains highly challenging. Such environments are often chaotic, unstructured, and hazardous, frequently giving rise to risk-conflict scenarios, in which environmental uncertainties interact with internal system inconsistencies (Clerigues et al., 2024; Granadeno and Cleland-Huang, 2025) . However, existing studies rarely provide systematic analyses of these scenarios or examine how they influence operators' perception, cognitive workload, and adaptive decision-making. As a result, a comprehensive understanding of risk-conflict elements and their effects on human–swarm interaction in SAR operations remains limited.

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In high-risk and time-critical missions, operators' situational awareness (SA)—the ability to perceive, comprehend, and project the state of the environment and relevant entities (Endsley, 1988, 1995)—plays a central role in mission planning and decision-making. However, environmental complexity, conflicting information, and large volumes of multisource data can degrade SA and increase cognitive workload. Closely related to SA is the resilience of human–swarm systems, which includes both system-level resilience, referring to the ability to maintain acceptable performance under disturbances through robustness, adaptability, and recoverability (Hollnagel et al., 2006), and operator-level psychological resilience, defined as the capacity to remain stable and adaptive under pressure (Feder et al., 2009). In human–machine collaboration, these two forms of resilience are interdependent: improved SA and adaptive automation can reduce cognitive load and stress, thereby supporting operator resilience (Fletcher and Sarkar, 2013), while resilient operators are better able to manage system capabilities, strengthening the overall resilience of the human–swarm system.

Although resilience engineering has been explored in intelligent systems, its application to UAV-swarm-based HSI in SAR missions remains limited. Existing studies primarily focus on improving the resilience of individual subsystems, such as communication networks, formation control, and task allocation, while resilience at the level of the human–swarm collaborative system remains relatively underexplored. Moreover, swarm resilience is highly context dependent, and current HSI frameworks emphasize coordination and information sharing but provide limited support for stable collaboration under risk-conflict conditions.

Meanwhile, as SAR environments become increasingly dynamic and information-rich, traditional teleoperation interfaces and two-dimensional graphical user interfaces provide limited support for perception and decision-making. Recent advances in Virtual, Augmented, and Mixed Reality technologies offer new opportunities to enhance human–swarm interaction by integrating spatial information into immersive environments (Coovert et al., 2014; Nagi et al., 2014; Bruce et al., 2017; Walker et al., 2018; Macchini et al., 2019; Rosen et al., 2020; Zheng et al., 2023; Catellani et al., 2024; Morando et al., 2025). In SAR scenarios, Mixed Reality interfaces can help operators better perceive the operational environment and understand swarm behavior (Catellani et al., 2024; Morando et al., 2025). However, research on MR-based resilient interaction design for UAV swarm SAR remains limited, particularly regarding its effects on situational awareness, cognitive workload, and psychological resilience under risk-conflict conditions.

To address these gaps, this study introduces MR technology to create a more immersive operational environment for UAV swarm SAR. The goal is to enhance operator situational awareness, reduce cognitive workload, and strengthen psychological resilience through resilience-oriented interaction design. Based on this research context, the following research questions are proposed:

RQ1. What types of risk-conflict scenarios arise in UAV swarm-based SAR missions, and how do these scenarios affect operators' cognitive and perceptual abilities?

RQ2. What are the interface design strategies for HSI in risk-conflict scenarios specific to UAV swarm-based SAR missions

RQ3. Can resilient design in MR HSI system enhance operator SA and psychological resilience in risk-conflict scenarios?

To address RQ1, this study identifies risk-conflict factors in UAV-based wilderness SAR operations and analyzes the characteristics of joint perception and cognition in human–swarm interaction. Based on situational awareness theory, a resilience interaction framework for human–swarm collaboration is proposed. For RQ2, typical risk-conflict scenarios were derived through questionnaires and expert interviews. The operational needs of UAV operators were analyzed, leading to the development of an MR-based HSI interface design framework. Interface prototypes were iteratively refined through usability testing. For RQ3, a human–swarm rescue simulator was developed based on the proposed framework. Experimental evaluation demonstrates that the resilient MR HSI system reduces operators' cognitive workload while improving situational awareness and psychological resilience in risk-conflict SAR scenarios.

2. Related works

2.1. Human swarm interaction of UAV

Human–Swarm Interaction (HSI), an extension of Human–Robot Interaction (HRI), focuses on communication and coordination between human operators and large groups of autonomous agents such as UAVs or ground robots (Kolling et al., 2015). Unlike traditional HRI, which typically involves one-to-one or small-scale interaction, HSI requires operators to supervise distributed autonomy and interpret the collective behavior of multiple agents. HSI in UAV swarms has attracted increasing attention due to its potential in applications such as emergency response, search and rescue (SAR), surveillance, and environmental monitoring (Kolling et al., 2015; Hoang et al., 2023; Siean et al., 2023). Human participation enhances system flexibility and supports more reliable decision-making in swarm operations (Kolling et al., 2013; Divband Soorati et al., 2021).

In UAV-based SAR missions, operations often occur in risk–conflict scenarios, where environmental uncertainties interact with internal system inconsistencies (Clerigues et al., 2024; Granadeno and Cleland-Huang, 2025). These conditions introduce multiple layers of unpredictability and require close coordination between human operators and autonomous agents. Within such environments, risks mainly originate from external factors such as extreme weather, unstable communication links, or sensing delays that reduce system reliability and situational awareness (Sende et al., 2024) . Conflicts, in contrast, emerge within the human–swarm system due to resource competition, inconsistent goals, or divergent decisions among agents (Guo et al., 2024; Ewers et al., 2025; Zhao et al., 2025). If unmanaged, these risks and conflicts may trigger cascading failures and weaken system resilience.

From a cognitive perspective, these conditions introduce substantial challenges for human operators. Compared with traditional HRI, HSI requires operators to monitor many autonomous agents simultaneously while maintaining global situational awareness and responding to dynamic mission conditions. This process involves continuous shifts between strategic and tactical thinking, complex attention management, and dynamic trust calibration with autonomous systems (Zhou et al., 2019; Bobko et al., 2023).

These challenges become more pronounced in risk–conflict scenarios. Operators must cope with environmental uncertainty, time pressure, and competing objectives among human and robotic agents. The large volume of information and urgency of decision-making significantly increase cognitive load, which may undermine situational awareness and disrupt trust calibration.

To address these challenges, existing research has progressed along two main directions: swarm control mechanisms and immersive interaction paradigms.

The first direction focuses on control mechanisms for managing swarm behavior. Research has explored strategies such as interaction modalities and operation techniques(Coovert et al., 2014; Liu and Shen, 2020; Fletcher et al., 2022), path planning methods (Angelopoulos et al., 2022), information sharing mechanisms (Reardon et al., 2019), and navigation or interaction techniques (Salunkhe et al., 2025) .Recent work has also incorporated computational cognitive models to support operator decision-making. For example, Gu et al. (2023) proposed a runtime probabilistic modelling and verification framework that

predicts swarm mission outcomes, improving anticipatory situational awareness. Building on this work, Abioye et al. (2024, 2025) demonstrated that Predictive Formal Modelling (PFM) can improve task efficiency and resource management in HSI.

The second direction explores immersive interaction environments based on Virtual, Augmented, and Mixed Reality (VAM). Traditional teleoperation interfaces and two-dimensional graphical user interfaces often fail to support the perceptual and cognitive demands of swarm operations. Interfaces integrated with VAM technologies can enhance human–machine collaboration by providing immersive representations of the operational environment (Nagi et al., 2014; Bruce et al., 2017; Walker et al., 2018; Macchini et al., 2019). They also improve the visualization of collaborative information and help communicate robot intentions more clearly to human operators (Coovert et al., 2014; Rosen et al., 2020).

Immersive environments strengthen spatial perception and support situational awareness, enabling more timely decision-making (Langerwisch et al., 2013). Consequently, recent studies have explored Mixed Reality (MR) interfaces for complex applications such as search and rescue and wildland firefighting. For example, Walker et al. (2021) developed an MR system to support ground robots in emergency response missions, while Salunkhe et al. (2025) designed an MR interface for UAV navigation tasks. Other studies have also developed mixed virtual–physical swarm platforms and MR-based interaction techniques for UAV operations (Angelopoulos et al., 2022; Chae et al., 2023; Zheng et al., 2023).

Despite these advances, existing HRI theoretical frameworks remain insufficient for human–swarm collaboration. Models such as the Supervisory Control Model (Sheridan and Verplank, 1978), Situation Awareness Theory (Endsley, 1999), and the Levels of Autonomy paradigm (Zhou et al., 2019) were developed for single-robot or small-scale interactions. They do not fully capture the distributed coordination, cognitive complexity, and emergent dynamics of swarm systems (Lewis, 2013; Hocraffer and Nam, 2017).

Although research on UAV swarm operations for wilderness SAR has made notable progress in interface and visualization design, several critical research gaps remain:

- **Limited understanding of operator cognition in risk–conflict scenarios.** Existing studies rarely examine how cognitive load dynamically evolves when operators face simultaneous external risks and internal conflicts. This limits our understanding of adaptive decision-making, attention allocation, and trust calibration in high-stress environments.
- **Inadequate interaction frameworks for swarm systems under uncertainty.** Current frameworks derived from traditional HRI fail to capture the nonlinear, distributed, and emergent characteristics of human–swarm coordination, highlighting the need for models that incorporate resilience, adaptability, and shared situational awareness.
- **Lack of resilient MR interaction strategies.** Although MR environments can enhance perception and coordination, few studies explore how such interfaces mitigate cognitive overload and stabilize situational awareness during high-risk missions.

To address these gaps, this study investigates how immersive MR interaction interfaces can support operators' situational awareness and task coordination during risk–conflict SAR missions, providing guidance for resilient human–swarm interaction design.

2.2. Situation awareness

Situation Awareness (SA), defined by (Endsley, 1988), refers to an operator's ability to perceive, comprehend, and project the state of the environment and the behavior of relevant entities. Building on this concept, Endsley (1995) proposed the widely adopted three-level SA model consisting of perception, comprehension, and projection. Subsequent research expanded this framework to include individual SA, team SA, and shared SA in collaborative environments (Salas et al., 1995; Stanton et al., 2017).

In search-and-rescue (SAR) operations involving UAV swarms, SA emerges not only from individual operators or agents but also from multi-level information exchange within the system. Such coordination enables a shared understanding of mission-relevant situations and supports collaborative decision-making under uncertainty. Consequently, swarm coordination often reflects characteristics across individual, team, and system-level SA (Stanton et al., 2017; Gao et al., 2018).

Recent studies have begun extending SA frameworks to multi-agent and UAV-swarm environments. (Han et al., 2024) proposed a multi-agent SA framework consisting of three layers—situation-element identification, current-state comprehension, and advantage assessment—and demonstrated its effectiveness in a SAR UAV-swarm case study. Gao and Li (2020) introduced a consensus-based distributed collaboration approach grounded in SA theory, enabling UAV agents to autonomously infer current and future environmental states through an observe–orient–decide–act cycle. Similarly, Jones et al. (2020) proposed a distributed SA concept for multi-robot systems, allowing robotic teams to collectively interpret environmental states and respond accordingly.

These advances highlight the growing importance of distributed and system-level SA in swarm operations. However, less attention has been paid to how human operators maintain situational awareness when supervising UAV swarms under conditions of uncertainty, high temporal pressure, and complex information flows. Empirical studies indicate that SA degradation is closely associated with reduced human performance, increasing the likelihood of misjudgment and operational errors (Fernandez Rojas et al., 2019).

Within the HSI domain, emerging work has begun to explore this issue. Kapellmann-Zafra et al. (2016) examined human interaction with simulated robot swarms under degraded SA conditions and found that limited SA significantly constrains operator performance. Wattearachchi et al. (2025a) further incorporated SA principles into HSI interface design and evaluation frameworks, combining insights from human–robot interaction and swarm intelligence (Bashyal and Venayagamoorthy, 2008). These studies provide valuable theoretical foundations for interface design in dynamic environments.

Nevertheless, existing research mainly focuses on general human–machine collaboration scenarios and provides limited guidance for high-risk SAR operations. In particular, there is still insufficient understanding of how HSI interfaces can support operators in maintaining SA while managing large volumes of information and coping with dynamic environmental uncertainties.

Overall, although SA modeling for UAV swarms and distributed systems has progressed substantially, important gaps remain regarding how interaction interfaces can effectively support operator SA in complex SAR scenarios. In such environments, operators face significant cognitive demands and are vulnerable to SA degradation. Therefore, this study investigates how HSI interface design can enhance operator SA in risk–conflict situations, improve information perception and comprehension efficiency, and reduce cognitive workload.

2.3. Resilience of HMC

Resilience originally emerged from ecology, describing a system's ability to maintain integrity under disturbance (Ludwig et al., 1997) . It has been further defined as the capacity to anticipate risks, respond to stress, and recover from disruptions ((Westrum, 2017). Core characteristics of resilient systems include risk anticipation, state monitoring, flexible response, and experiential learning (Steen and Aven, 2011). Disruptions may arise from technical failures, human errors, or environmental factors. In human–machine collaboration, resilience supports workload reduction and enhances system performance.

In the field of Human-Machine Collaboration (HMC), Fong et al.

(2003) define HMC as an interaction paradigm in which humans and robots make joint decisions toward shared objectives, treating both parties as collaborative partners. Amalberti (2006) introduced a resilience model dividing system regulation into a predict–detect–recover cycle (Fig. 1-a), emphasizing anticipation, monitoring, and recovery. Hollnagel et al. (2006) proposed the resilient organization model (Fig. 1-b), highlighting continuous monitoring of system indicators to maintain operational performance under unexpected events. Building on this, Zieba et al. (2010) developed an adaptive behavior model (Fig. 1-c), which addresses disturbances at both agent and system levels using prediction, diagnosis, and trial-and-error reasoning. Recent extensions refine these frameworks for autonomous socio-technical systems, incorporating counterfactual reasoning and adaptive policy synthesis to support agent-level adaptation (Samarasinghe, 2023; Johansson and Lundberg, 2025).

In these models, system adaptation occurs at two levels. Agent-level (local) resilience adjusts parameters within the current operational mode to maintain functionality. System-level (global) resilience modifies operational modes themselves to achieve broader recovery. These levels are progressive: local resilience forms the foundation for global resilience, which integrates multi-agent coordination for overall system optimization.

However, existing resilience models assume stable system dynamics and continuous access to system states—conditions rarely met in UAV swarm–based SAR operations. Wilderness rescue environments are highly uncertain, with discontinuous information flows and rapidly evolving threats. Moreover, current models often overlook operators' cognitive and psychological dimensions, even though situational awareness and psychological resilience critically shape overall human–swarm system performance. These gaps limit the applicability of classical resilience frameworks in high-risk SAR scenarios.

Current UAV swarm resilience research mainly addresses local swarm components (e.g., communication, network coverage, obstacle avoidance, path planning) and system-level performance (Clerigues et al., 2024). Human–machine collaboration studies highlight the benefits of human–robot teaming for disturbance management (Flood and Keegan, 2022; Tyler et al., 2025) but remain largely technology-centered, neglecting operators' psychological resilience—defined as the ability to maintain stability and adapt under stress (Feder et al., 2009). Interface transparency, task allocation, and automation level directly affect operators' stress, cognitive load, and decision reliability (Fletcher and Sarkar, 2013).

Recent work has explored integrating frameworks such as HEFFR and FMD-based tools for resilience modeling in complex HMI systems (Irshad and Hulse, 2022), demonstrating conceptual feasibility. However, these approaches remain largely unvalidated in high-risk, domain-specific contexts like UAV-swarm SAR and insufficiently capture situational dynamics, such as escalating threats and degraded information quality.

To address these gaps, this study develops a resilience-oriented framework for HSI in UAV-swarm SAR, integrating both system-level and operator-level resilience. Building on this framework, resilient Mixed Reality (MR) interface strategies are proposed and empirically evaluated via simulation-based SAR experiments. The study examines how resilience-oriented interaction design affects operators' cognitive load, situational awareness, and psychological resilience under risk-conflict scenarios.

3. Resilient interactive design for human-swarm SAR

3.1. Analysis of cognitive elements in HSI

3.1.1. Risk conflict analysis in SAR

In the context of HSI for wilderness SAR operations, rescue teams face challenges and risks stemming from environmental, technical, and human factors. The ability of intelligent systems to maintain critical functions, performance, and user experience in the face of various challenges, pressures, failures, or unexpected situations is of paramount importance. Therefore, based on the rescue scenario, an HSI framework has been developed, as shown in Fig. 2. By analyzing the risk-conflict scenarios in swarm-based rescue operations from the perspectives of the rescue environment, tasks, operators, and the system, it becomes easier to systematically understand the composition of risk-conflict scenarios in wilderness SAR operations.

(1) From the perspectives of the environment and tasks, environmental complexity and task urgency may introduce risks for HSI. Wilderness SAR operations are typically conducted in natural environments, where complex, uncertain, and unstructured terrain conditions may impede rescue progress and increase the time and difficulty of operations. In addition, the rescue environment may be affected by adverse weather conditions, such as heavy rain, snowstorms, or high temperatures. Rescue missions are often complex and multi-stage, including planning, reconnaissance, and execution. During execution, teams face significant time pressure. In emergency situations, rescuers may lack sufficient time for comprehensive risk assessment and planning, which can lead to hasty decisions or operational errors. Time constraints may also result in uneven resource allocation, leaving critical tasks insufficiently supported. Prolonged high-intensity work may further lead to fatigue, which can impair judgment and reaction time.

(2) From the perspective of the rescue personnel, rescuers' perception and cognitive abilities are influenced by both the environment and task demands, which may introduce additional risks. In high-pressure situations, rescuers may experience cognitive biases, such as overconfidence or confirmation bias, which can affect decision quality. In complex and unpredictable rescue

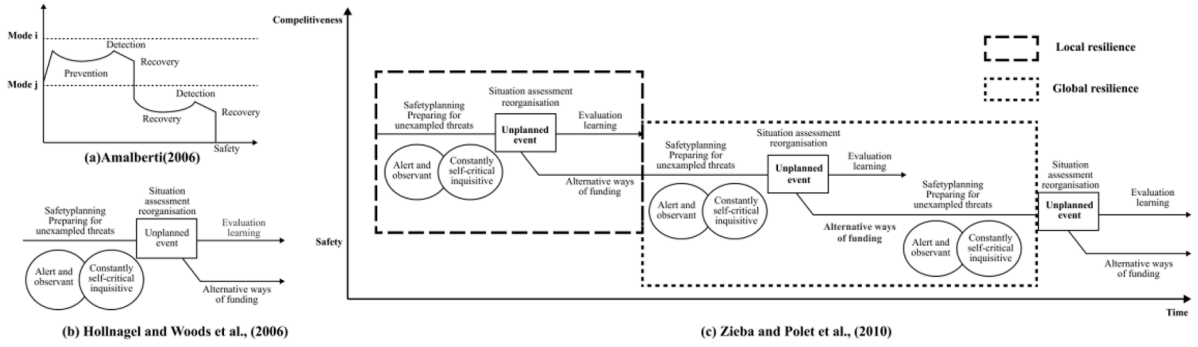


Fig. 1. Resilience Model Proposed in Previous Studies. (a) Amalberti (2006) proposed the predict–detect–recover regulation cycle.; (b) Hollnagel and Woods (2006) introduced the resilient organization model based on continuous monitoring and recovery; (c) Zieba and Polet (2010) extended this framework into a dynamic feedback process enabling adaptive coordination.

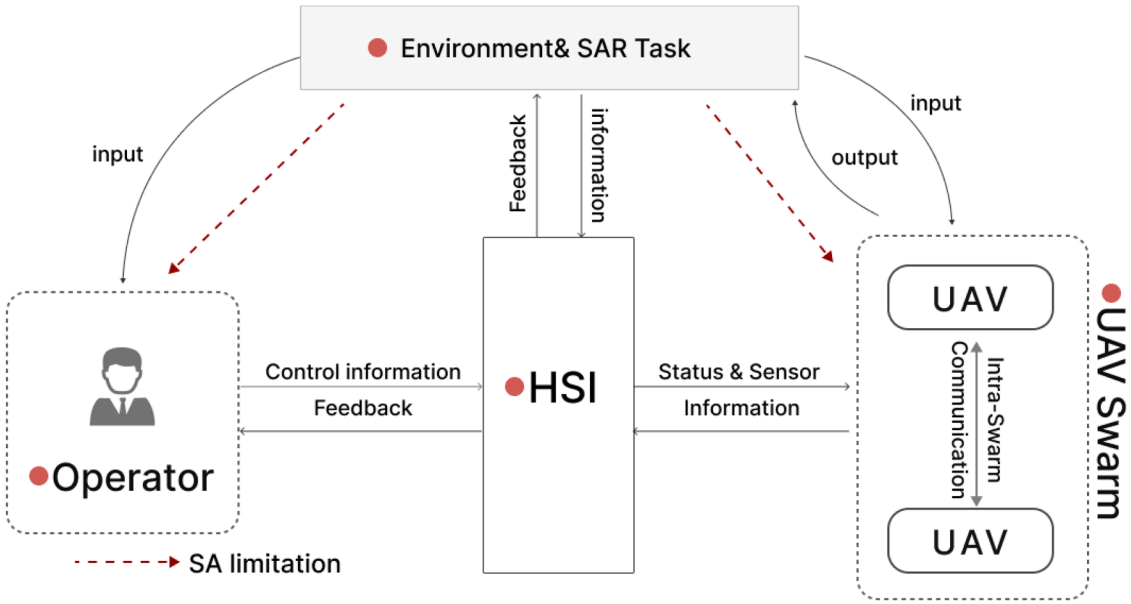


Fig. 2. Human–swarm collaborative interaction model in wilderness SAR scenarios. Operators and UAV swarms form a collaborative HSI team, exchanging information via the interface to support coordination and joint decision-making. Solid arrows indicate primary information flows, dashed arrows indicate environmental influences. Red dots denote risk sources from environmental, operator, and UAV factors that may affect situational awareness and coordination.

environments, situational uncertainty may exceed the operators' perceptual capacity, increasing the likelihood of missing critical information. At the same time, urgent rescue tasks often impose high cognitive workload on operators. Unexpected or emergency events during the mission may also create psychological stress for rescue personnel.

(3) From the perspective of the rescue system itself, in hazardous environments, the UAV system may experience task failures or reduced efficiency due to hardware or software malfunctions. According to prior studies, common system-level failures in swarm-based SAR include:
- Communication loss or signal degradation, which disrupts data exchange among UAVs or between the swarm and the operator, potentially leading to task interruption or swarm fragmentation (Lucchesi et al., 2024).
- Sensor or navigation failure, such as GPS signal occlusion, visual sensor damage, or IMU drift, which may compromise environment perception and navigation accuracy (Phadke and Medrano, 2022; Zhang and Liu, 2023).
- Battery depletion or propulsion malfunction, which can result in individual UAV crashes or premature withdrawal from the mission, forcing resource redistribution and increasing operator cognitive load (Alqudsi and Makaraci, 2025).

These failures often interact with environmental and cognitive risks to form risk–conflict loops—for instance, a UAV's battery failure may trigger communication congestion and mission replanning, heightening the operator's cognitive burden and degrading situational awareness.

In summary, from the perspective of the human-machine-environment interaction in outdoor rescue, the risk conflicts that may arise during UAV swarm rescue operations mainly stem from the complexity of the environment, the rescue tasks, the UAV system's performance, and the capabilities of the operators themselves.

3.1.2. Collaborative cognitive interaction framework for HSI

In the face of risk conflicts, the resilience of the HSI system helps predict environmental changes, adapt to these changes, and restore the system's normal functioning under such circumstances. Human users, through proactive coping strategies and adaptability, can effectively respond to challenges, make decisions, and continue executing tasks. The classical SA model emphasizes an individual's ability to perceive, comprehend, and predict environmental states. However, it does not explicitly consider resilience capabilities, such as responding, adapting, and recovering, which are essential for human–machine systems operating under risk conflicts.

Therefore, based on the three-layer SA model, this study introduces the concept of resilience, integrating resilience dimensions (adaptation, recovery), and constructs a joint cognitive framework with five layers of cognitive structure (Perception—Comprehension—Prediction—Adaptation—Recovery), as shown in Fig. 3**.** Combining the resilience organizational model (Amalberti, 2006) and resilience regulation behavior model (Zieba et al., 2010), in this paper, the diagnostic and trial-and-error reasoning phases in the resilience model are defined as "adaptation." The behavior model of human-machine systems facing risk conflict events is defined as the three main activity phases of Prediction—Adaptation—Recovery.

During wilderness SAR missions, the operator, UAV swarm, and HSI resilience system form a dynamic and interdependent collaborative system. The UAV swarm continuously monitors the environment and analyzes risk information to support operator decision-making and maintain mission execution. When a risk-conflict event occurs, the system first collects and analyzes relevant information, identifies the event, and predicts its potential evolution. It then provides explanatory information to help operators maintain situational understanding under pressure and support adaptive responses. Finally, the system offers operational suggestions based on the evolving situation. These suggestions facilitate psychological recovery and enable operators to continue decision-making and task execution.

In this interaction model, the resilience-enhanced interaction system improves the operator's psychological resilience in response to risk-conflict events through a Prediction—Adaption—Recovery mechanism, thereby enhancing the overall resilience of the HSI system. This framework strengthens resilience through three core mechanisms:

(1) **Human-Swarm Joint Cognition:** Operators and UAV swarms achieve coordinated decision-making by sharing situational awareness through the HSI interface. The interface encodes different types and levels of risk information, ensuring that

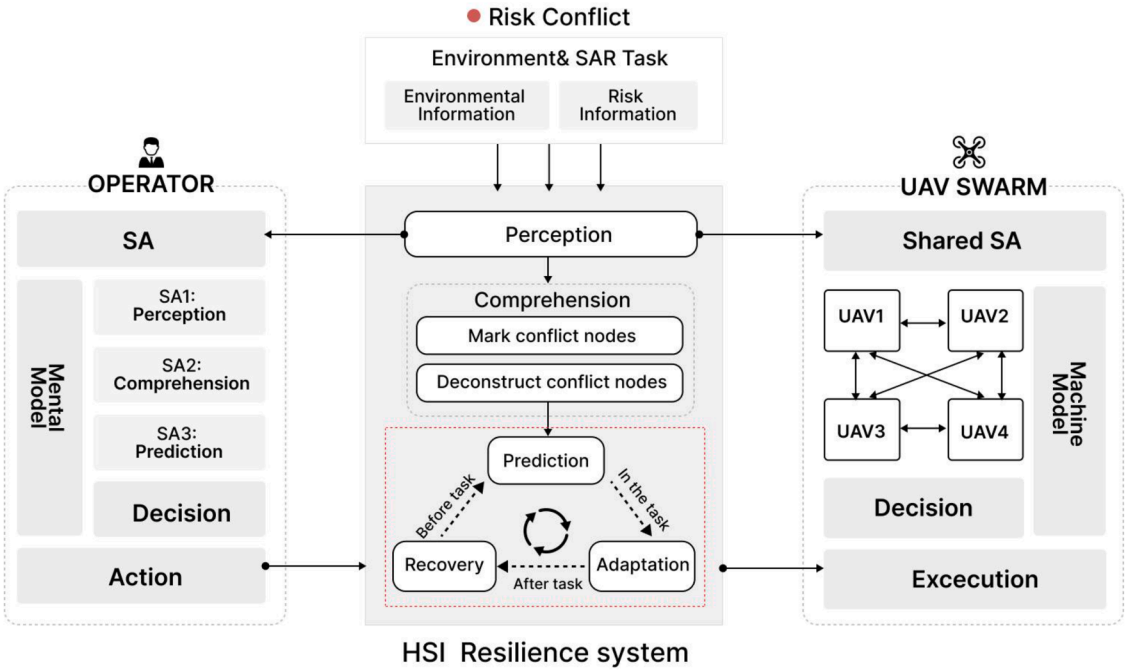


Fig. 3. SA-based HSI resilience interaction framework. The figure presents a resilience-oriented HSI framework centered on shared situational awareness (SA). The HSI resilience system functions as an intermediary cognitive layer between the human operator and the UAV swarm, supporting the construction and sharing of SA during risk–conflict scenarios. Environmental and task information is processed through a five-layer cognitive structure—perception, comprehension, prediction, adaptation, and recovery—which enables a continuous predict–adapt–recover regulation cycle. This mechanism facilitates adaptive human–swarm coordination and enhances both operator psychological resilience and overall system resilience in dynamic SAR environments.

operators can continuously monitor the status of the UAV swarm (e.g., health status, path planning, task allocation) as well as environmental and personnel dynamics (e.g., survivor locations, casualty information) in real time.

(2) **Situational Awareness—Resilience Support Mechanism:** The HSI system enhances the situational awareness of both the operator and the UAV swarm, improving their ability to perceive environmental information, identify potential risk-conflict situations, and better prepare for and adapt to these challenges.

(3) **Dynamic Coordination of Prediction, Adaptation, and Recovery:** When the system identifies a risk-conflict event, it analyzes and deconstructs the event, marks key nodes, and supports the operator's decision-making through resilient system design. Initially, the system provides risk assessments and early warning information, helping the operator understand the current level of risk and potential consequences. When a risk materializes, the system guides the operator and the UAV swarm to take appropriate response measures and operational actions, thereby mitigating the impact of the risk and facilitating rapid recovery to normal operations.

3.2. Design of HSI resilience interactive interface

3.2.1. User research and needs exploration

The objective of this phase of the study is to address RQ2 by employing user research methods to explore, from the users' perspective, the needs of operators interacting with the HSI system during wilderness SAR missions when facing risk-conflict events. The aim is to identify corresponding design opportunities for the HSI interface.

To investigate these questions, the study first conducted a desktop research review to map out the key personnel involved in human-swarm wilderness SAR operations, including on-site technical staff, back-end data analysts, team leaders, expert groups, and medical rescue personnel. Furthermore, through focus group discussions, the study developed a relationship diagram of key roles in the wilderness SAR scenario, as shown in Fig. 4.

Next, to develop potential solutions, semi-structured interviews were conducted with experts to gather information and feedback regarding domain-specific interface design. The participants included 3 experts with practical experience in UAV-assisted SAR operations and 4 general users.

Through the semi-structured interviews, it was summarized that operators encounter various risk-conflict events during actual SAR missions. Among these, communication failures of UAVs were identified as the most urgent and performance-critical risk-conflict event across different emergency scenarios. Therefore, this event was selected as the representative use case for this study. Subsequently, a cognitive work analysis was conducted to map out the primary task flow for the study, as shown in Fig. 5.

The basic requirements for MR HSI system and the operational pain points encountered during actual rescue missions were summarized and consolidated. Based on this analysis, three main categories of system user functions and information needs were identified. Furthermore, several key functionalities were determined through which the MR HSI system could enhance emergency response operations during wilderness SAR.

3.2.2. Design framework for HSI resilience interface

In order to translate the user needs collected into the interface design of HSI interaction system, this section proposes an HSI interface design framework for risk-conflict scenarios, as shown in Fig. 6. The goal of the framework is to guide information processing and presentation across the three stages of resilience design.

According to information processing theory, human cognition involves sequential stages of input, processing, storage, and output. Similarly, in risk-conflict scenarios, HSI systems undergo stages of information input, processing, and presentation. The proposed framework maps these stages onto the five-layer cognitive architecture of the joint cognitive resilience model, aiming to enhance operators' situational awareness (SA) and psychological resilience through multi-level

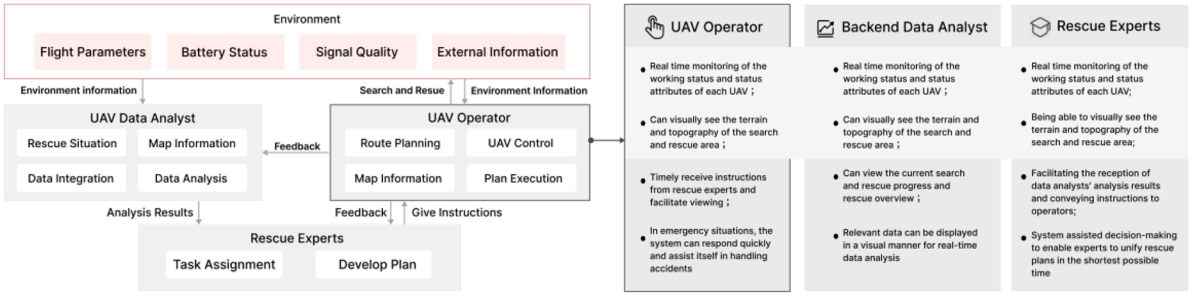


Fig. 4. Role Relationship Diagram and User Requirements. This framework illustrates the information flow and functional relationships among key stakeholders in UAV swarm search and rescue missions. Environmental parameters—such as flight status, battery level, signal quality, and external information—are monitored to support decision-making. UAV data analysts integrate mission and environmental data to provide insights for rescue experts and UAV operators. Rescue experts develop rescue plans and issue task instructions, while UAV operators perform route planning, UAV control, and task execution. The right section summarizes the user requirements of three core roles—UAV operators, backend data analysts, and rescue experts—highlighting their needs for real-time monitoring, data visualization, situational awareness, and coordinated decision-making.

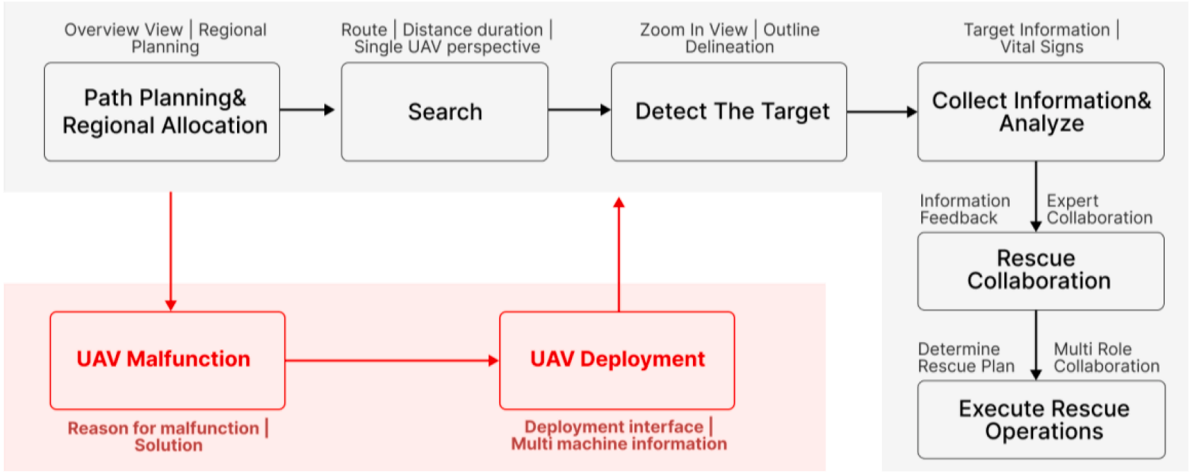


Fig. 5. Workflow of UAV swarm-based SAR missions and system functions. The diagram presents the operational workflow of UAV swarm–based search and rescue missions. The process includes path planning, multi-UAV collaborative search, target detection, and information analysis, followed by expert-supported rescue decision-making. The lower section highlights two key resilience mechanisms: UAV malfunction management, which supports diagnostic feedback and recovery actions, and UAV redeployment, which enables adaptive task reallocation and coordinated mission continuation in dynamic environments.

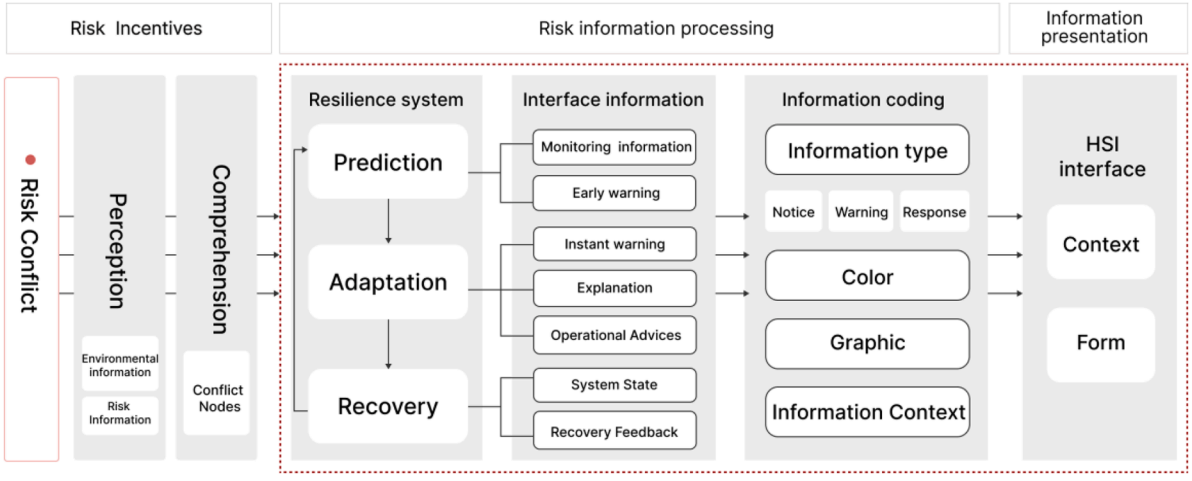


Fig. 6. HSI Interface Design Framework. This framework illustrates the information encoding process of a resilient HSI system in risk-conflict scenarios. It comprises three stages: risk incentivization, risk information processing, and information presentation. In the first stage, the system perceives and understands risk stimuli. In the second stage, it encodes the information using predictive-adaptive-recovery attributes. In the final stage, the processed information is presented to the human–swarm team via the HSI interface.

information encoding and dynamic feedback. The prediction, adaptation, and recovery layers represent the core resilience attributes of the interface. The functional mappings of this framework at each cognitive layer are as follows:

At the input stage, the framework primarily acts on the perception and comprehension layers of the interface. Environmental and risk information serve as stimuli to the entire rescue system. Upon receiving these stimuli, the system's sensors capture and transmit the information to the operating system, which subsequently identifies and processes the received data.

(1) **Perception Layer:** The UAV swarm uses sensors, such as environmental sensors and UAV status monitors, to collect real-time environmental data. These data include survivor locations, rescue area boundaries, and UAV positions. The system also monitors operator states, such as attention level and fatigue.
(2) **Comprehension Layer:** The HSI system processes data collected in the perception layer to construct an integrated understanding of the rescue environment and tasks. When risk-conflict events occur, the system labels and decomposes them for further analysis. During the processing stage, the system extracts key risk elements and encodes risk information. Resilient design is applied to three interface layers: prediction, adaptation, and recovery.
(3) **Prediction Layer:** Based on the current SA, such as UAV status, battery levels, mission paths, and allocated areas, the system predicts potential future risks and anticipates task demands. In the interface, the prediction attribute is represented through monitoring information and early warning alerts. These may include notifications about weather changes, terrain obstacles, or communication disruptions. This layer emphasizes the system's ability to anticipate risks and monitor environmental changes, which helps operators respond effectively to risk-conflict events.
(4) **Adaptation Layer:** The HSI system provides dynamic adjustment strategies to respond to environmental changes. Examples include reallocating UAV tasks, supporting collaborative decision-making, and adjusting operational actions. In the interface, this layer presents disaster warnings, operational details, and explanatory information about event causes. These functions help operators understand emergencies and select appropriate response strategies. The adaptation layer highlights the system's ability to adjust to evolving conditions.
(5) **Recovery Layer:** The system supports rapid recovery after task failures or conflicts. It guides operations back to the original mission trajectory and maintains mission continuity. In the interface, the recovery attribute is implemented through actionable suggestions and feedback on system status restoration. This layer focuses on restoring normal operations after disruptions. It supports operators by providing feasible action plans. Operators can then apply their professional knowledge to make informed decisions and continue rescue operations.

Based on the above, to achieve efficient information transmission, a layered coding strategy is adopted. The information types are categorized and prioritized as follows:

(1) **Notifications:** Low-priority information (such as task progress updates) is presented with neutral colors and static icons to avoid distracting from the main tasks.
(2) **Warnings:** These are used to highlight high-risk (e.g., UAV battery below 20%, UAV failure, discovery of survivors) and emergency events (e.g., discovery of survivors). They are indicated by a red flashing effect combined with spatialized alarm sounds to alert operators of critical events.
(3) **Responses:** Instructions requiring operator confirmation (e.g., "Confirm") are presented in high-contrast colors with reminder boxes, combined with audiovisual feedback (e.g., "Successfully returned") to ensure timely response and decision-making.

For visual encoding, different colors represent different information types. Notifications and responses use bright green and blue to indicate system feedback. Warning information uses vivid red. Different shades of red represent varying urgency levels. The system divides warnings into four levels. Low, medium, and high intensities of red correspond to increasing risk frequency and impact on user experience. Universal and simple icons are used for warning symbols (Fig. 7), allowing rescue personnel to identify critical information quickly.

Based on the task scenarios and corresponding interface properties, the warning messages should be designed clearly to indicate the cause of the event and provide actionable suggestions. The design elements of the interface should incorporate the information encoding features (such as color, symbols, and alerts) to guide the operators efficiently.

The interface prototype should be developed according to the resilience design framework and the interface design principles derived from user research. After the prototype is created, multi-stage usability testing should be conducted to ensure alignment with users' mental models and to evaluate the user experience. The results will help identify design issues and support iterative improvements. After testing, a high-fidelity interface prototype will be finalized to support the development of a simulator for further evaluation.

4. Experiment design

4.1. Objectives and hypotheses

The experimental scenario was set in a mountainous wilderness area where a flash flood trapped climbers in a valley. Heavy rainfall made roads impassable, preventing ground rescue teams from reaching the site. Consequently, a UAV swarm was deployed to conduct search operations and provide critical situational information to support rescue teams.

Based on this scenario, this experiment investigates how resilience-oriented human–swarm interaction (HSI) design influences operators' situational awareness (SA), cognitive load, and psychological resilience when encountering risk–conflict events.

Situational awareness (SA) refers to the perception, comprehension, and projection of environmental elements (Endsley, 1995) . Maintaining SA in dynamic environments requires effective integration and presentation of multi-source information. Mixed Reality (MR) interfaces can overlay geo-referenced and task-relevant data—such as UAV locations, victim positions, and hazard zones—directly within the operator's spatial view, reducing the need for mental translation between displays and the physical environment.

Recent human–swarm interaction studies indicate that MR interfaces can improve SA by providing spatially aligned and context-aware visualizations in multi-agent environments (Billinghurst and Duenser, 2012; Catellani et al., 2024; Lyons et al., 2025; Morando et al., 2025) . However, their effectiveness in risk-conflict rescue scenarios remains underexplored. Therefore, we propose:

H1. In risk-conflict scenarios, the MR HSI interface can effectively enhance the operator's SA.

According to Cognitive Load Theory (Sweller, 1988), optimizing the presentation of information can reduce extraneous cognitive load. During UAV swarm supervision, operators must process large volumes of dynamic information under time pressure, leading to increased visual search effort and working-memory demand.

MR interfaces can reduce these demands by presenting information in a spatially integrated and centralized manner, thereby decreasing visual search effort and context switching. Previous studies have reported reduced cognitive workload in MR environments compared with traditional 2D interfaces (Merriënboer and Sweller, 2010; Zheng et al.,

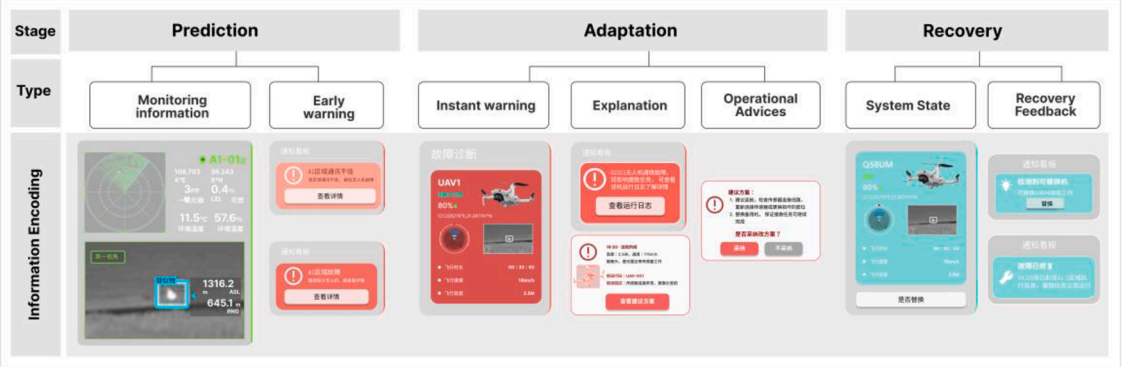


Fig. 7. Interface Reminder Information Design. The figure depicts the interface information types and information design in the three resilience design stages of prediction adaptation recovery. The prediction stage mainly includes monitoring information and early warning information, the adaptation stage includes instant warnings, explanations, and operational advices, and the recovery stage includes system status and recovery feedback information. And assign different colors to the interface information based on the severity of the situation, with red indicating danger, green indicating safety, and blue indicating the recovery status from danger emergency to safety.

2023; Suzuki et al., 2024; Hou et al., 2025) . Therefore, we propose:

H2. In risk-conflict scenarios, the MR HSI interface can effectively reduce the operator's cognitive workload.

Psychological resilience refers to the ability to maintain effective performance and adapt under stress. Resilient systems design emphasizes supporting operators' cognitive processing and reducing uncertainty in high-risk environments (Hollnagel, 2014).

Studies in XR and human–robot interaction indicate that interface transparency and in-situ information presentation can influence users' trust, emotional stability, and acceptance of system information (Navas-Medrano et al., 2024; Taub et al., 2024) . Compared with AR, MR provides deeper spatial integration between virtual and physical elements, which may further support situational understanding and emotional regulation in dynamic environments. Therefore, we propose:

H3. In risk-conflict scenarios, an MR HSI interface with resilient attributes can enhance the operator's psychological resilience.

4.2. Participants

Prior to the main experiment, a pilot study with 6 participants was conducted to verify the experimental procedure, task difficulty, and reliability of data collection. Based on the observed effect sizes, a post hoc power analysis was performed using G*Power to estimate the required sample size.

Given the within-subject design with two conditions (control and experimental interfaces), a paired-samples *t*-test was selected. With a significance level of $\alpha = 0.05$ (two-tailed) and statistical power of 0.80, the analysis indicated that at least 34 participants were required to detect a medium effect size (Cohen's $d = 0.5$). Therefore, 35 participants were recruited to ensure sufficient statistical power while accounting for possible data loss or noise in physiological recordings.

This sample size is consistent with previous HSI and MR-based studies that used similar participant numbers (Kolling et al., 2013; Abbas and Jeong, 2024; Wattearachchi et al., 2025b) . The final sample of 35 participants was therefore considered adequate for the experimental objectives.

Participants were recruited through the university platform and the researchers' professional networks. The sample included students and graduates from engineering-related disciplines as well as professional UAV operators. Among them, 5 participants had prior experience in UAV-assisted search-and-rescue (SAR) operations, while 32 participants reported general UAV operation experience.

All participants had normal or corrected-to-normal vision and no history of motion sickness. Preference was given to individuals familiar with UAV interfaces and VR headsets to reduce learning effects. In total, 30 participants reported prior experience with VR headsets. This composition allowed evaluation of system usability and cognitive performance across both experienced operators and novice users.

Participants' ages ranged from 18 to 35 years ($M = 28.8$), including 15 males and 20 females. The experiment was conducted in a controlled laboratory environment with ethical approval from the university's review board. All participants provided informed consent and received monetary compensation. Before the formal experiment, participants completed 10–15 min of guided familiarization with the simulation system. Each experimental session, including post-task interviews, lasted approximately 50–65 min.

4.3. Experimental equipment and materials

4.3.1. Experimental environment

We developed a multi-agent human-machine rescue simulation program using Unity software, incorporating two interface design schemes based on the high-fidelity prototypes and principles for MR interface design outlined above. Using the formative feedback and research work collected in Section 3, we developed a simulation system for UAV swarm in SAR. This simulator operates on VR devices and creates a virtual environment within the system to simulate the display environment in MR settings. The overall layout of the interface follows the basic dimensions and structure of MR interface development, with the interactive interface designed according to the task flow. The two developed interface design schemes were designated as the control group and the experimental group in this study.

The layout of the interface in the simulator is shown in Fig. 8. The MR interface can be divided into three levels: the permanent interface layer, the surrounding layer, and the mixed reality layer.

In the permanent interface layer, as it remains in the user's field of view for an extended period, the interface information deployed at this level includes: (1) Key Metrics: For example, normalized system monitoring indicators. (2) Control Panel: Simulates physical 3D input devices, utilizing the 3D functionality of VAM displays, such as manipulable control rings or floating control areas.

Based on the MR interface and the existing functional information architecture, the following information functional architecture is derived based on the MR interface layers:

First Layer: The operational interface and navigation bar are fixed and permanently visible within the operator's field of view, not changing positions with the operator's movement. This facilitates easier interaction and presentation of information.

Second Layer: This layer features an interactive block sandbox. The operator can switch the monitoring perspective to a third-person view of

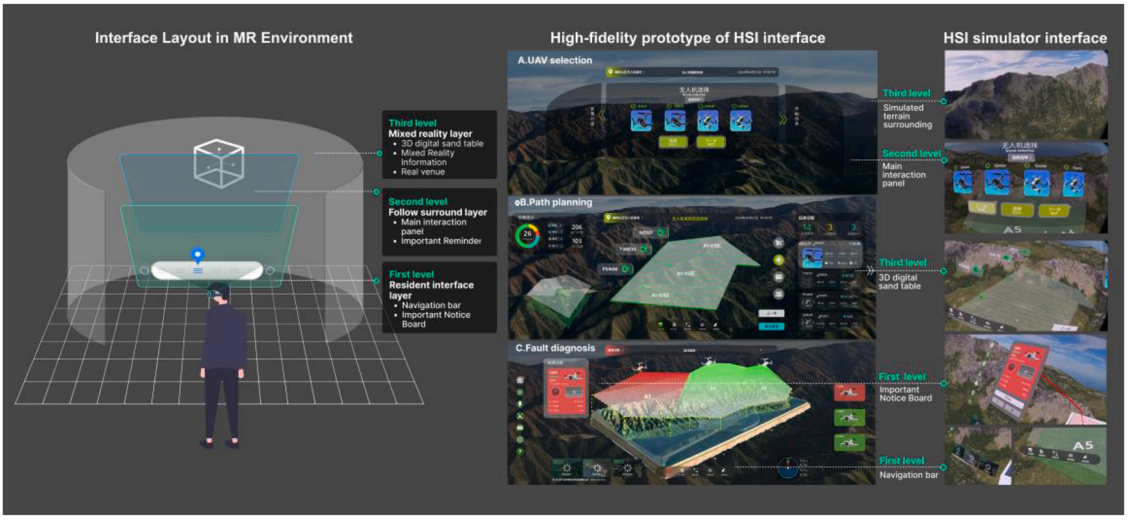


Fig. 8. MR interface layout and HSI interface layout. The figure illustrates the hierarchical structure of the MR-based HSI simulation environment and representative interface prototypes used in the experiment. The MR interface is organized into three layers: the resident interface layer, which embeds virtual content within the simulated environment; the follow-surround layer, which contains spatially anchored interaction panels; and the mixed reality layer, which presents critical system information and alerts. Panel (A) shows the UAV selection interface integrated with the terrain environment. Panel (B) presents the path-planning interface using a spatial sandbox for trajectory adjustment. Panel (C) illustrates the fault-diagnosis interface, where alert and status panels are placed in the MR layer to ensure visibility during time-critical operations.

the sandbox, allowing them to observe the exact location and detailed information of survivors. The sandbox is rotatable and scalable.

Third Layer: The third layer of the MR environment is based on 3D terrain, simulating an immersive outdoor rescue environment.

4.3.2. *Experimental equipment*

The experimental equipment used in this experiment is shown in Fig. 9.

The research utilizes the Oculus Quest HMD as the display device, and the Side Quest software is used to monitor the operator's actions via video.

For physiological data monitoring, the study employs a physiological data collection wristband developed by the Chinese Academy of Sciences. This wristband can collect physiological data such as ECG and GSR, helping researchers analyze the emotional and physiological changes of participants when they encounter unexpected situations during the operation. This device supports the synchronized collection of various physiological signals and behavioral data, and it allows for both automatic and manual tagging of specific events during the experiment. After the experiment, the data is automatically synced to the cloud, where researchers can download raw and feature data for the analysis and evaluation of emotional states, psychological stress, fatigue

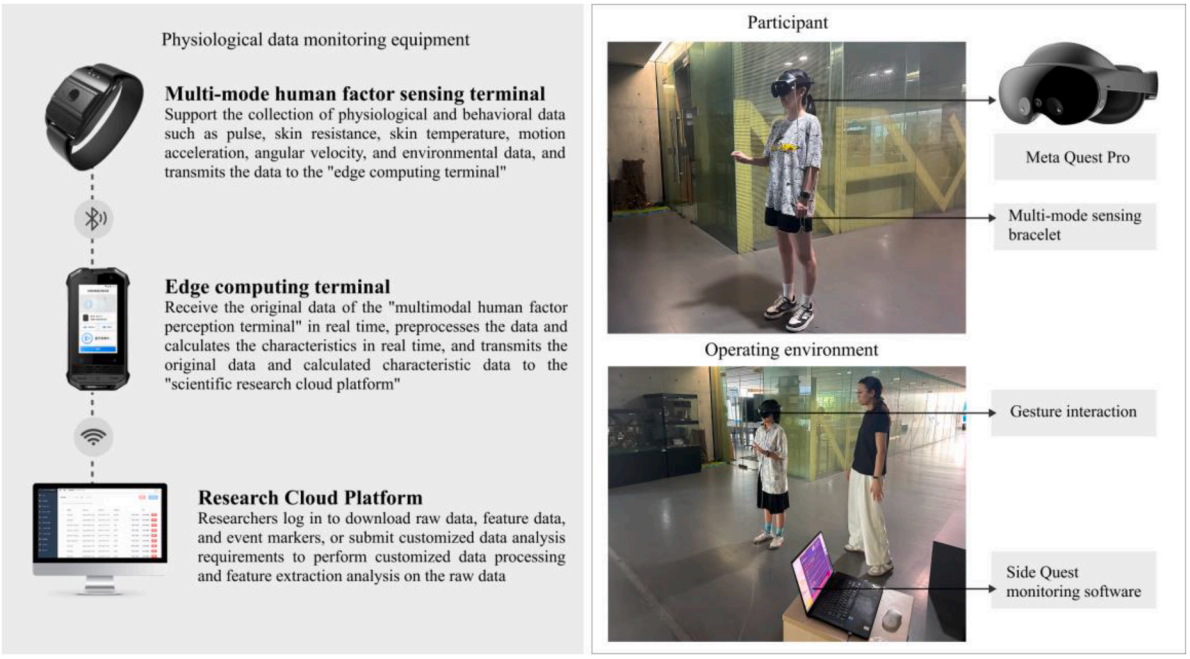


Fig. 9. Experimental equipment and operating environment. The figure shows the Meta Quest Pro HMD used for gesture-based interaction and the physiological monitoring setup, which includes a multimodal wristband connected to an edge computing terminal. Side Quest software records participants' VR views and interaction behaviors for subsequent analysis.

levels, and cognitive load, among other psychological conditions.

4.4. Procedure

The experimental process consists of the following steps:

(1) Pre-Experiment Training: Before the formal experiment begins, participants undergo relevant training, which includes an introduction to the rescue background, equipment operation instructions, and details regarding post-experiment questionnaires and interviews. This ensures that participants are familiar with the operational procedures and safety protocols.
(2) Physiological Data Collection: Participants are equipped with physiological devices to collect data during the experiment.
(3) Formal Experiment: Participants are instructed to wear the Quest headset and complete the two experimental scenarios. During the process, they are encouraged to engage in think-aloud protocols, openly verbalizing their thoughts and challenges encountered during the tasks. After completing the scenarios, participants fill out the experiment questionnaire.
(4) Post-Experiment Interviews and Feedback: After the formal experimental tasks, participants are interviewed to gather their feedback, understand their views on the experiment, and collect suggestions for improvement.
(5) Result Compilation: Data collected during the experiment, including UAV operation data, MR technology interaction data, and rescue effectiveness evaluation data, is organized. Questionnaire results are analyzed, and interview feedback is summarized and integrated into the overall findings.

4.5. Task settings

According to the task flow, the UAV operator needs to use the rescue system to control the UAV swarm, completing tasks such as area allocation, autonomous patrolling, and path planning. After receiving the search and rescue task, the operator begins searching for survivors. Once a potential survivor is detected, the operator must send feedback to the rescue team and decide whether to proceed with further rescue actions. At the same time, the operator must handle risk conflict events during the rescue process, such as dealing with UAV malfunctions, and continue to execute the search and rescue mission despite these challenges.

To reproduce risk–conflict events in the simulator, a representative UAV failure—sensor connection anomaly—was triggered after a potential survivor target was detected within the task sequence. When the event was triggered, the UAV's visual feed in the lower-left corner corresponding to the affected UAV's camera view displayed a "connection lost" status for 5 s, indicating the UAV was offline. Participants could check the details, after which the system automatically diagnosed the failure and displayed the UAV fault. The system then provided a remediation solution by assigning a replacement UAV, whose corresponding camera view showed "reconnecting" until normal operation resumed. Although the failure occurred after the target search stage in the simulation settings, participants were unaware of the exact timing, making the event effectively unpredictable and preserving the realism of the risk–conflict scenario.

4.6. Experimental variable design

The experiment adopts a single-factor repeated measures design. The experimental variable consists of the interface designs corresponding to the prediction, adaptation, and recovery attributes, representing the three attributes of resilience design. The comparison chart in Fig. 10 explains the attributes of the interface design corresponding to the experimental group.

Control Group: HSI Interface without Resilience Attributes (Perception-Comprehension-Prediction)

(1) Reminder Panel Design: When a UAV malfunction occurs during the target search, the system provides the fault reason and solution for the operator's reference. The operator adjusts the interface, views multi-UAV information, and completes the UAV allocation and replacement to continue the rescue mission.
(2) Perspective Thumbnail: In the absence of faults, the system presents normal monitoring, with the border color and label in

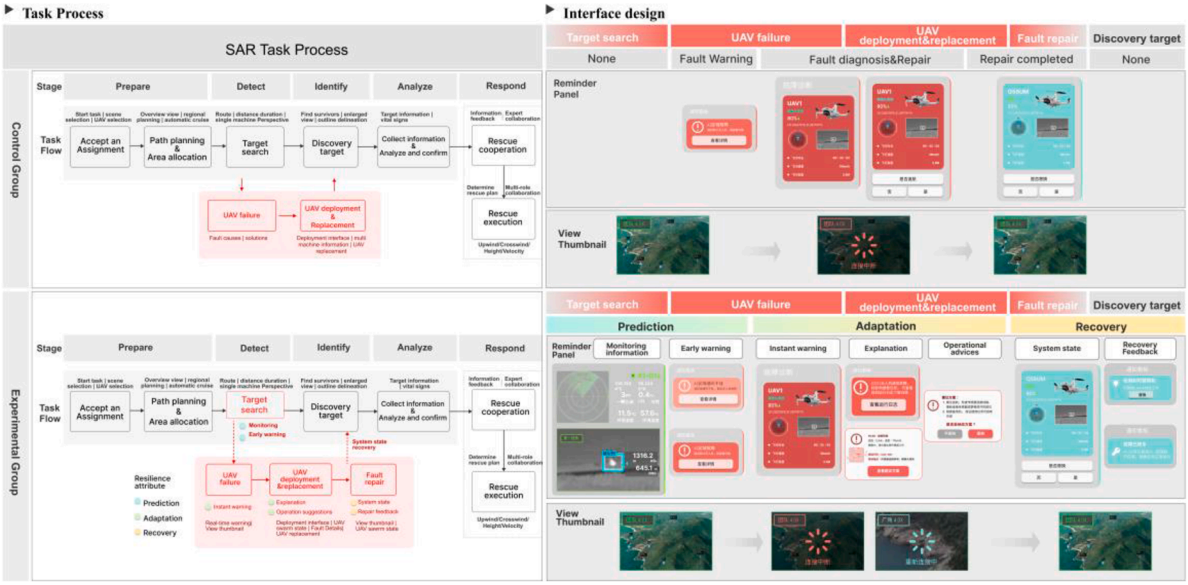


Fig. 10. Task flow and interface design of experimental and control groups. The left panels illustrate the task sequence for both groups, including the phases "Prepare," "Detect," "Identify," "Analyze," and "Respond." In the control group, UAV failures are handled through basic fault warnings, UAV deployment or replacement, and fault repair interfaces. The right panels show the corresponding simulator interfaces for each task stage. The experimental group incorporates additional resilience-oriented interface elements in three categories: prediction, adaptation, and recovery. Prediction interfaces provide monitoring information and early warning panels. Adaptation interfaces deliver instant alerts, explanatory information, operational guidance, and a thumbnail view indicating system adaptation status. Recovery interfaces present system state information and recovery feedback.

green. Upon fault occurrence, a red symbol indicating connection interruption is displayed, and the border turns red. After the fault is resolved, the system restores the original display state.

Experimental Group: HSI interface with resilience attributes (Perception-Comprehension-Prediction-Adaptation-Recovery)

(1) Reminder Panel Design: During the target search phase, monitoring information is displayed with early warning of risk conflict events. In the event of a UAV malfunction, the system provides a disaster warning, labels the fault reason, and the operator can view detailed fault causes and issues. The system then provides operational suggestions for the operator, who can decide to replace the UAV. Once the UAV replacement is completed and the fault is resolved, the system sends feedback indicating the repair is complete, allowing the operator to continue the rescue mission.
(2) Perspective Thumbnail: In the absence of faults, the system presents normal monitoring, with the border color and label in green. Upon fault occurrence, a red symbol indicating connection interruption is displayed, and the border turns red. After the UAV is replaced, the label and symbol change to blue, indicating that the system is reconnecting. Once the fault is repaired, the system restores to its original state.

The experimental group's interface incorporates additional resilience-oriented elements across three categories: interfaces with predictive attributes, interfaces with adaptive attributes, and interfaces with recovery attributes. The interfaces with predictive attributes provide monitoring information and early warning panels, perceiving environmental data and identifying potential threats. Upon detecting a possible risk, the system issues a warning to alert the operator about the likelihood of the risk occurring. The interfaces with adaptive attributes deliver immediate alerts, explanatory information, and operational guidance. When risk conflicts cannot be resolved automatically, such as during a UAV malfunction, the system provides suggested solutions and countermeasures, allowing the operator to decide whether to implement the solution, replace the UAV, and repair the malfunction. The interfaces with recovery attributes present system state and recovery feedback. If the solution is implemented, the human-machine coordination system restores its original functionality; otherwise, the system performance may degrade (e.g., transitioning from a cooperative two-UAV search to a single-UAV operation).

In the prediction phase, the MR HSI system monitors system status and data, perceives environmental data, and identifies potential threats. Upon detecting a possible risk, the system issues a warning to alert the operator about the likelihood of the risk occurring. If the risk is mitigated, the warning is considered successful.

When the risk conflict within the system cannot be resolved, such as a UAV malfunction, the interaction system provides the operator with solutions and countermeasures. The operator decides whether to implement the solution, replace the faulty UAV, and repair the malfunction. If the solution is not implemented, the performance of the human-machine coordination system will degrade (for example, transitioning from a cooperative search operation with two UAVs to a single UAV search). If the solution is implemented, the human-machine coordination system will restore its original functionality.

4.7. Dependent variable

(1) Situational Awareness (SA): The operator's understanding and prediction ability of the environment are assessed through situational awareness tests in the simulated tasks. The SART scale (Taylor, 2017) is used, with a 7-point Likert scale. Each level of the interface corresponds to a SART scale, with ratings ranging from 1 (very low) to 7 (very high).
(2) Cognitive Workload: The NASA Task Load Index (NASA-TLX) (Hart and Staveland, 1988) is used to measure the workload during system operation. Higher scores indicate higher demands. A 5-point Likert scale is employed.
(3) System Usability: System usability is evaluated using the System Usability Scale (SUS) (Bangor et al., 2009), which assesses the usability of the interface, with items 1, 6, and 9 specifically used to measure satisfaction. A 5-point Likert scale is used, ranging from 1 (strongly disagree) to 5 (strongly agree).
(4) Resilience: The resilience is evaluated using the CD-RISC Connor-Davidson Resilience Scale (Connor and Davidson, 2003) and the self-resilience scale (ER89) (Farkas and Orosz, 2015). The CD-RISC is a unidimensional self-report scale containing 25 items designed to measure psychological resilience. It uses a 5-point Likert scale, ranging from "not at all true" = 0 to "true nearly all the time" = 4, with higher scores indicating greater psychological resilience. The total score is the sum of individual item responses.

    The ER89 scale is a unidimensional self-report scale with 14 items designed to measure trait resilience. Responses range from "not at all applicable to me" = 1 to "completely applicable to me" = 5.
(5) Response Time: Response time is measured using time metrics proposed by Zieba et al. (2010) which includes evaluations of unexpected events. The resilience evaluation includes time metrics such as T-transition, T-detection, T-switch and T-decision, as shown in the diagram below.

    T-detection: It refers to the time interval between the occurrence of an unexpected event and the determination of the need for adjustment. In this experiment, the time from the occurrence of a fault event to the inspection and diagnosis of the fault is calculated.

    T-switch: The time required to effectively switch between two modes. Calculate the return time of the UAV for fault diagnosis in this experiment.

    T-decision: It is to set an appropriate adjustment time and calculate the time for replacing the faulty UAV after returning in this experiment.

    T-transition is the sum of T-detection, T-switch, and T-decision indicators.
(6) Physiological data indicators:Physiological measurements have been widely employed in the field of human-computer interaction due to their sensitivity and accuracy in detecting mental workload (Hao et al., 2022). Such measurements are effective for assessing mental workload, which refers to the demands placed on an individual's limited cognitive resources to achieve the desired level of task performance. These demands can impose pressure on an individual's attention, emotions, and responses (Meshkati and Loewenthal, 1988). Excessive mental workload can negatively affect the efficiency and reliability of human-machine systems (Hao et al., 2022).

A considerable number of studies have utilized physiological signals, such as GSR, HR (Healey and Picard, 2005; Singh et al., 2013), EEG (Ma et al., 2021; Gao et al., 2022), and HRV (Isler and Kuntalp, 2007; Xu et al., 2014), to conduct valuable investigations into mental workload.

The analysis of HRV indicators mainly falls into three categories: time-domain, frequency-domain, and nonlinear metrics (Bhardwaj and Balasubramanian, 2019). Among these, time-domain and frequency-domain analyses are the most common linear methods used to evaluate HRV. In this study, we employ selected HRV metrics to evaluate the impact of the interaction design schemes on participants' physiological states.

The metrics used in this study include RMSSD, pNN50, and SDNN, as shown in Table 1. These indicators are closely associated with parasympathetic nervous system activity and can reflect participants'

Table 1
Physiological Data Indicators. The table below describes the units and basic descriptions of the four metrics (RMSSD, pNN50, SDNN, and HR) detected in this experiment.

Parameter	Units	Description
RMSSD	ms	The root means square value of the difference between adjacent RR intervals
pNN50	%	Ratio of the NN50 to the number of RR intervals
SDNN	ms	The standard deviation (SD) of the RR intervals
HR	bpm	The heart rate

relaxation levels and physiological recovery states. Under high cognitive workload conditions, parasympathetic activity typically decreases, leading to a reduction in these indices.

RMSSD: RMSSD measures the square root of the mean squared differences between adjacent RR intervals and reflects short-term heart rate variability. It is particularly sensitive to rapid modulation of heart rate by the parasympathetic nervous system. pNN50: pNN50 calculates the proportion of successive RR interval differences greater than 50 milliseconds, indicating the extent of larger variations in heartbeat intervals. It reflects the overall strength of parasympathetic regulation and may be associated with longer-term or more pronounced changes in autonomic balance.

SDNN: SDNN represents the overall variability of heart rate. With increasing workload, a decrease in SDNN typically indicates a shift toward sympathetic nervous system dominance, reflecting heightened physiological stress.

HR: HR refers to the number of heart beats per minute (bpm) and serves as a key indicator of physiological arousal and autonomic nervous system activity. In stressful contexts, HR generally increases due to sympathetic activation, reflecting an elevated physiological tension. Conversely, a lower and more stable HR typically indicates better autonomic regulation and emotional stability.

4.8. Result and analysis

4.8.1. Scale results

To evaluate participants' performance, perception, and psychological responses across experimental conditions, a series of standardized scales were administered following each task. All participants (N = 35) completed each scale immediately after both the control and experimental sessions, ensuring within-subject comparability.

The Shapiro-Wilk test confirmed that all measured variables followed a normal distribution. Accordingly, paired-sample *t*-tests were employed for subsequent analyses. Data visualization was performed using Prism software, with significance levels indicated as follows: * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p \leq 0.0001$.

As shown in Fig. 11, the results of the paired-sample *t*-test revealed a significant difference between the experimental group and the control group ($t(34) = 4.279$, $p < 0.0001$), indicating that the experimental group demonstrated superior performance in enhancing operator SA compared to the control group.

Subsequently, further paired-sample *t*-tests were conducted on the three dimensions of the SART scale, as illustrated in Fig. 11(a). The results showed that the experimental group exhibited a significant reduction in attentional demand, whereas no significant differences were found in attentional supply ($t(34) = 1.098$, $p = 0.2798$) and understanding ($t(34) = 1.576$, $p = 0.1243$), as shown in Fig. 11(h). These findings suggest that the experimental interface primarily enhanced operators' awareness of situational instability and variability, thereby improving their ability to comprehend the complexity of the environment and thus boosting situational awareness. However, the system did not show a significant effect on improving operators' attention allocation or information understanding.

Cognitive Workload: As shown in Fig. 11(b), the results of the paired-sample *t*-test revealed an extremely significant difference between the two groups in terms of cognitive workload ($t(34) = 5.190$, $p < 0.0001$), indicating that the experimental group performed better in reducing operators' cognitive workload compared to the control group.

Fig. 11(g) also presents the results of further paired-sample *t*-tests

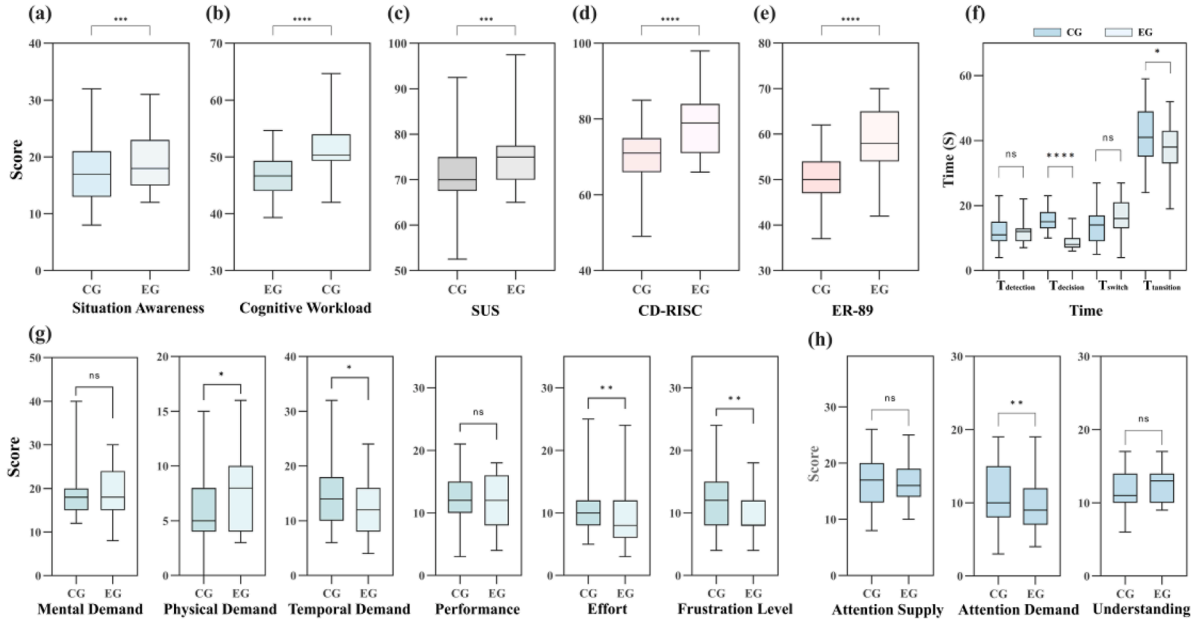


Fig. 11. Scale data results. Fig. 11(a)–(e) present boxplots and paired statistical test results comparing the control group (CG) and the MR-based experimental group (BG) in Situation Awareness (SA), Cognitive Workload, System Usability Scale (SUS), Connor–Davidson Resilience Scale (CD-RISC), and ER-89. Higher scores indicate better performance for SA, SUS, CD-RISC, and ER-89, whereas lower scores indicate lower workload for the Cognitive Workload scale. Fig. 11(f) presents boxplots and paired test results for time-related task performance. Fig. 11(g) and 11(h) show results for the six dimensions of the Cognitive Load Scale and the three dimensions of SA, respectively. Statistical significance is indicated as ns (not significant), $p < 0.05$ (*), $p < 0.01$ (**), and $p < 0.0001$ (***).

across the six NASA-TLX dimensions. The results showed that the experimental group exhibited significant improvements in Temporal Demand, Performance, Effort, and Frustration Level. However, Mental Demand and Physical Demand were higher in the experimental group compared to the control group, with the difference in Mental Demand being non-significant ($t(34) = 5.190$, $p < 0.0001$). Participants reported lower temporal, physical, and mental demands, as well as reduced effort and frustration levels, while achieving significantly higher performance scores, suggesting that participants perceived their task performance to be better under the experimental interface.

The main findings are as follows:

(1) Significant differences ($p < 0.05$) were observed between the two groups in Physical Demand and Temporal Demand, with the experimental group reporting higher scores in Physical Demand and the control group reporting higher scores in Temporal Demand.
(2) Greater significance ($p < 0.01$) was found in Effort and Frustration Level, both showing that the control group scored significantly higher than the experimental group.
(3) No significant differences ($p > 0.05$) were found between the two groups in Mental Demand and Task Performance.

System Usability: As shown in Fig. 11(c), the results showed that the system usability scores for the experimental group's interface were significantly higher than those of the control group ($t(34) = 4.240$, $p < 0.001$), indicating a highly significant difference in system usability between the two groups. According to the SUS results, the final usability score for the experimental group was 85.125, suggesting a very high level of usability based on the six-level SUS grading system.

Psychological Resilience: As shown in Fig. 11(d) , the results of the paired-sample *t*-test indicated a statistically significant difference between the experimental group and the control group in the CD-RISC psychological resilience scale scores ($t(34) = 6.314$, $p < 0.0001$). The experimental group consistently scored higher than the control group, suggesting that the introduction of the experimental interface may help enhance operators' psychological resilience and adaptability when facing risk and conflict.

Similarly, the results for the ER89 scale, as shown in Fig. 11(e) , revealed a highly significant difference between the experimental and control groups ($t(34) = 12.69$, $p < 0.0001$). Participants in the experimental group demonstrated significantly better emotional recovery capabilities compared to the control group, indicating that the augmented reality system not only improved the driving interaction experience but also facilitated faster emotional recovery from disturbances, thereby enhancing driving stability and safety.

Transition Time: As shown in Fig. 11(f), the differences in response times between the control group (CG) and the experimental group (EG) across three phases—event detection, task switching, and decision-making—after encountering a risk conflict event are presented. In the T-decision phase (i.e., the time taken to make a replacement decision after completing fault diagnosis), the experimental group exhibited significantly faster response times compared to the control group ($t(34) = 10.63$, $p < 0.0001$), indicating that the introduction of resilience features in the interface notably enhanced operators' decision-making efficiency under high-pressure conditions, reduced cognitive load, and accelerated task progression.

In the T-detection phase (i.e., the time from the onset of the risk conflict event to its identification), although the experimental group showed slightly better median times than the control group, the difference was not statistically significant ($t(34) = 0.2827$, $p > 0.05$). This suggests that the resilient interface design had limited impact on the initial risk perception phase, possibly because this stage relies more on the operator's instinctive reactions or system warning stimuli.

Similarly, in the T-switch phase (i.e., the time taken to switch control modes or operational strategies after fault identification), the difference between the two groups was also not significant ($t(34) = 1.194$, $p > 0.05$), implying that response times during this stage are mainly constrained by system response and interface structure, with limited influence from resilience attributes.

Considering the overall transition response time, represented by T-transition, the experimental group outperformed the control group significantly ($t(34) = 2.489$, $p < 0.05$). This finding further demonstrates that although the resilient interface did not exhibit significant advantages in every single stage, it contributed to higher overall task response efficiency across the multi-stage task sequence.

4.8.2. Correlation analysis of resilience dimensions

To further examine the internal relationships among resilience-related factors and their contribution to overall resilience, a correlation analysis was conducted based on the revised CD-RISC scale. The dataset comprised valid responses from 35 participants, each completing the modified scale under both control and experimental conditions. All questionnaire data were preprocessed to remove missing values and normalized to a [0,1] range prior to analysis to ensure comparability across dimensions.

Given that the variables were ordinal and not strictly interval-scaled, Spearman's rank correlation coefficient (ρ) was employed to evaluate the monotonic relationships among the five resilience dimensions and the overall resilience index. Analyses were conducted separately for the control group and the experimental group to capture interface-induced variations in resilience structure. Statistical testing was performed using SPSS and GraphPad Prism, with significance levels defined as $p < 0.01$, $p < 0.05$*, and $p < 0.1$. In addition, to ensure reliability, all diagonal self-correlations were validated ($p < 0.001$), confirming the stability and internal consistency of the revised scale structure.

Based on the resilience model proposed in this study, the CD-RISC items were reorganized into five theoretically grounded dimensions—perception, comprehension, prediction, adaptation, and recovery—as described below. A further Spearman correlation analysis was conducted on the CD-RISC scale. Based on the resilience model proposed in the study, the original items of the CD-RISC were modified to align with the experimental context, categorizing them into five dimensions: perception, comprehension, prediction, adaptation, and recovery.

In the revised scale, the five dimensions assess the following: the perception dimension evaluates awareness of the environment and tasks, as well as self-assessment; the comprehension dimension assesses the ability to analyze environmental conditions, understand the meaning of tasks, teamwork, and one's own role; the prediction dimension evaluates anticipation of future risks and task demands along with confidence levels; the adaptation dimension assesses the flexibility and initiative to dynamically adjust strategies in response to changes; and the recovery dimension evaluates the ability to rebuild information, regulate emotions, and restore functionality after setbacks and difficulties.

Subsequently, Spearman correlation analysis was used to explore the relationship between each dimension and the final measured variable of resilience. The analysis was conducted separately for the control group and the experimental group, and the results are presented in Fig. 12.

All significance levels were marked in three tiers (***$p < 0.01$, **$p < 0.05$, *$p < 0.1$). Significance tests for the main diagonal self-correlations all passed validation ($p < 0.001$), indicating the statistical reliability of the dimensional correlation patterns identified in this study. The correlation coefficients between variables ranged from −0.115 to 0.862, overall forming a positive correlation network as expected. Among them, correlations of moderate strength or higher ($r > 0.4$) accounted for 38.5%, consistent with the typical distribution characteristics of multidimensional scale measurement indicators.

The results of the Spearman correlation analysis for the experimental group showed that all five dimensions were highly and positively correlated with resilience (all p-values < 0.01), with correlation coefficients ranging from 0.762 to 0.855. This indicates that each sub-

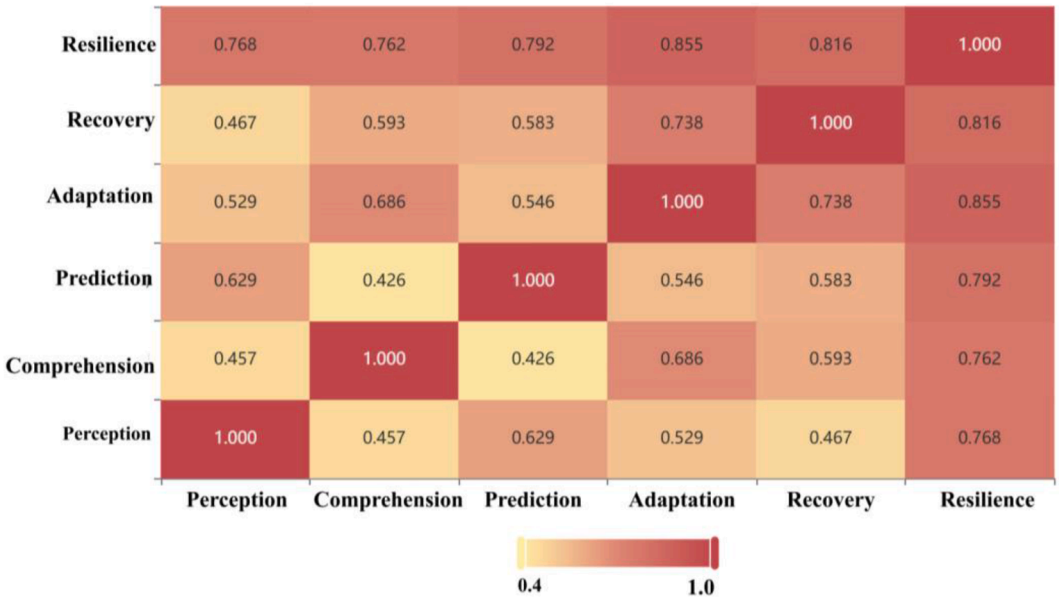


Fig. 12. Correlation analysis results heatmap. This figure presents the correlation analysis results among the five key resilience dimensions. The matrix shows the pairwise correlation coefficients, indicating the strength and direction of linear relationships between each dimension. The dimensions listed on both axes include Perception, Comprehension, Prediction, Adaptation, and Recovery. Colors range from yellow to red, with darker red indicating stronger positive correlations as coefficients approach 1.

dimension has a strong explanatory power for resilience. Pairwise correlation analyses among the dimensions revealed that all comparisons reached statistical significance ($p < 0.05$), with coefficients ranging from 0.426 to 0.738, suggesting that while the dimensions maintain independence, they are also interrelated.

The main findings include: (1) the adaptation dimension had the highest correlation with resilience (ρ=0.855), followed by the recovery dimension (ρ=0.816); (2) the strongest inter-dimensional correlation was found between adaptation and recovery (ρ=0.738, $p < 0.001$), followed by understanding and adaptation (ρ=0.686, $p < 0.001$); (3)the correlation coefficient matrix revealed that in the multidimensional structure, adaptation may play a central role, as its average correlation with other dimensions reached 0.625, higher than that of other dimensions; (4) the lowest inter-dimensional correlation was between understanding and prediction (ρ=0.426, $p = 0.011$), although it still represented a moderate strength association.

The results of the control group showed that all dimensions were significantly positively correlated with the overall resilience score ($p < 0.01$). The main findings are as follows: (1) the recovery dimension ($r = 0.862$) and the prediction dimension ($r = 0.810$) exhibited the strongest correlations with resilience; (2) the highest inter-dimensional correlation was between prediction and recovery ($r = 0.700$), indicating the most pronounced covariation between these two dimensions and suggesting that they contribute most significantly to the overall measurement; (3) prediction was also significantly positively correlated with perception ($r = 0.411$), understanding ($r = 0.448$), and recovery ($r = 0.700$), highlighting its central role within the scale; (4) the correlation between understanding and adaptation was not statistically significant (=0.144, $p = 0.409$), suggesting potential discriminant validity between these two constructs.

In summary, within the control group, prediction and recovery showed the strongest inter-dimensional correlation and the greatest covariation. Recovery also had the highest correlation with overall resilience, indicating its major contribution to the resilience assessment. It should be noted that these correlations reflect associative rather than causal relationships among resilience dimensions. Nevertheless, they provide evidence supporting the multidimensional construct validity of revised scales.

Compared with the control group, the experimental interface incorporated resilience attributes targeting the adaptation and recovery dimensions. Correlation analysis showed significantly stronger associations between these dimensions and overall resilience scores, supporting the effectiveness of the resilience-oriented interface design. Among all dimensions, adaptation demonstrated the strongest association with resilience, suggesting that interface features enhancing adaptive capabilities are particularly important for improving operators' psychological resilience.

These findings support the conceptual design model proposed in this study and provide practical implications for future human–AI interaction systems. In particular, interface design should place greater emphasis on supporting adaptive responses and post-event recovery, enabling users to respond more resiliently to dynamic or high-risk scenarios.

4.8.3. Physiological data analysis

Physiological data were continuously collected throughout the entire experimental process using a Multi-Mode Human Factor Sensing Terminal (as shown in Fig. 9). The terminal continuously recorded participants' physiological indicators, including heart rate (HR) and heart rate variability (HRV) parameters such as SDNN, pNN50, and RMSSD.

During data acquisition, an Edge Computing Terminal was employed to control the start, end, and calibration time points (T) of each recording segment, ensuring accurate synchronization between physiological data and experimental events. After each acquisition, raw physiological data were automatically uploaded to the Research Cloud Platform, which conducted the first stage of data preprocessing. This process included feature extraction, signal denoising, baseline correction, and the removal of incomplete or noisy segments caused by motion artifacts or sensor disconnections.

Following cloud-based preprocessing, the datasets were downloaded for centralized offline analysis using Prism software. Outliers exceeding three standard deviations from the mean were excluded, and all data were normalized to minimize inter-individual variability. After this multi-step cleaning process, 27 valid datasets were retained from the original 35 participant samples for subsequent analysis.

According to the experimental design, three key time points were

defined: T (fault occurrence), T-30 s (steady state), and T + 30 s (recovery phase).

Physiological indicators (SDNN, HR, pNN50, and RMSSD) were analyzed for both the control group (without resilience interface) and the experimental group (with adaptability and recovery resilience features). A paired samples *t*-test was performed to examine differences in physiological responses across the three time points within the experimental group. The detailed analysis results of physiological indicators are presented in Table 2 and Fig. 13.

(1) SDNN: At the steady state (T-30 s), no significant difference between groups was observed ($p > 0.05$). At the fault occurrence point (T), the experimental group exhibited significantly higher SDNN values compared to the control group ($t(26) = 0.2333$, $p < 0.05$). At the recovery phase (T + 30 s), the difference between the experimental and control groups remained significant ($t(26) = 1.145$, $p < 0.05$). These findings suggest that the resilience-enhanced interface (with adaptability and recovery features) improved participants' physiological regulation and recovery capabilities under stress, demonstrating stronger physiological resilience.

(2) HR: At both the fault point (T) and the recovery point (T + 30 s), the experimental group exhibited significantly lower heart rates compared to the control group ($p < 0.001$ and $p < 0.05$, respectively). This indicates that participants using the resilience-enhanced interface experienced lower arousal levels during fault conditions, suggesting that the interface effectively helped operators reduce physiological tension and cognitive load during high-stress tasks, thereby enhancing system recoverability.

Given that a notable difference was observed at the steady state (T-30 s), a further analysis was conducted to compare the change in heart rate (ΔHR) between T and T + 30 s. The experimental group demonstrated a mean HR decrease of $\Delta = -7.12\%$, whereas the control group exhibited an HR increase of $\Delta = +2.34\%$, indicating superior heart rate recovery in the experimental group.

(3) PNN50 and RMSSD: As shown in the figure, the experimental group consistently exhibited significantly higher PNN50 values at all three time points (T-30 s, T, and T + 30 s) compared to the control group ($p < 0.05$, $p < 0.01$, and $p < 0.001$, respectively). The highest level of significance was observed at the recovery phase (T + 30 s), reflecting stronger parasympathetic activation in the experimental group during stress exposure.

RMSSD: Similarly, significant differences in RMSSD were observed between groups at all three time points (T-30 s, T, and T + 30 s), with the experimental group consistently demonstrating higher values ($p < 0.05$, $p < 0.001$, and $p < 0.05$, respectively). Notably, a significant difference was already evident at the steady-state point (T-30 s), suggesting that the experimental group maintained better parasympathetic activation throughout both task preparation and execution phases.

Given that a notable difference was observed between groups at the steady-state point (T-30 s), a further analysis was conducted to examine the change (Δ) between the fault point (t) and the recovery point (T + 30 s) for two physiological indicators.

For pNN50, the experimental group exhibited a mean increase of $\Delta = +15.58\%$, whereas the control group showed a marked decrease of $\Delta = -27.17\%$. Similarly, for RMSSD, the experimental group demonstrated a mean increase of $\Delta = +21.72\%$, compared to a minimal increase of $\Delta = +2.69\%$ in the control group. These results suggest that the resilience-enhanced interface more effectively reduced cognitive load and promoted physiological recovery between the fault and recovery phases.

Overall, the physiological data further indicate that participants in the experimental group exhibited better emotional regulation and physiological recovery throughout the entire experiment, reflecting stronger psychological resilience and more stable operational performance.

4.8.4. User interview results (removed redundant wording and improved conciseness)

To evaluate the impact of the MR resilience interaction system on users' psychological resilience and stress tolerance in risk scenarios, semi-structured interviews were conducted following a structured qualitative analysis process. The interviews focused on participants' past experiences, system usage, psychological recovery, and design feedback. Data were collected through one-on-one semi-structured interviews lasting approximately 15 min each and audio-recorded with participants' consent. Recordings were transcribed verbatim and analyzed using qualitative content analysis to identify themes related to psychological resilience, interaction experience, and recovery behavior.

To ensure anonymity, interview excerpts are identified by participant codes (e.g., P01, P02). A total of 35 participants were interviewed, including 5 with prior UAV-based search-and-rescue (SAR) simulation or related project experience and 30 with general UAV operation experience. This composition allowed comparison between novice and more experienced operators. The interview guide and representative questions are provided in Appendix A to enhance research transparency.

Most participants reported that the 3D MR interface was more immersive and intuitive than traditional 2D systems. Perspective-switching functions were frequently mentioned as improving situational awareness (SA) and operational efficiency by enabling rapid transitions between global and local views. One participant noted, "*Compared to just clicking on a 2D screen, the 3D interface makes it easier to see what's happening around the UAV, and switching perspectives really helps me understand the situation faster*" (P07).

However, in high-stress situations, some participants reported cognitive overload caused by excessive visual information. For instance, one participant stated, "*There's just too much information popping up in this risk scenario… I don't have the mental capacity to read all the text while managing the UAV*" (P14). Others suggested simplifying visual outputs

Table 2
Results of physiological data analysis. Table 2 summarizes the statistical analysis of key physiological parameters, including SDNN, heart rate (HR), PNN50, and RMSSD, measured at three time points: 30 s before fault occurrence (T–30 s, steady state), at the moment of the fault (T), and 30 s after the fault (T + 30 s, recovery phase). For each parameter, the mean (M), standard deviation (SD), t-value (T), and significance level (P) from paired *t*-tests are reported.HR reflects overall physiological arousal, while SDNN, RMSSD, and PNN50 are heart rate variability (HRV) indicators associated with autonomic nervous system regulation. Higher HR values indicate increased arousal, whereas higher HRV-related values suggest stronger autonomic flexibility and stress regulation capacity. The results show significant physiological changes across the three phases, indicating dynamic stress responses and recovery following fault events. Statistical significance is denoted as $p < 0.05$ (**), $p < 0.01$ (**), $p < 0.001$ (***).

Parameter	T–30				T				T + 30			
	M	SD	T	P	M	SD	T	P	M	SD	T	P
SDNN	19.68	35.94	2.846	0.6477ns	1.277	28.45	0.2333	0.0343*	9.418	42.73	1.145	0.0473*
HR	5.482	8.359	3.408	0.0021 **	4.989	6.165	4.205,	0.0003 ***	−3.525	8.259	2.218	0.0355 *
PNN50	0.1254	0.2378	2.417	0.0253 *	−0.136	0.2186	2.859,	0.0097 **	0.1699	0.1829	4.256	0.0004 ***
RMSSD	14.12	32.84	2.235	0.0343 *	−11.49	14.95	3.993,	0.0005 ***	13.88	33.93	2.125	0.0432*

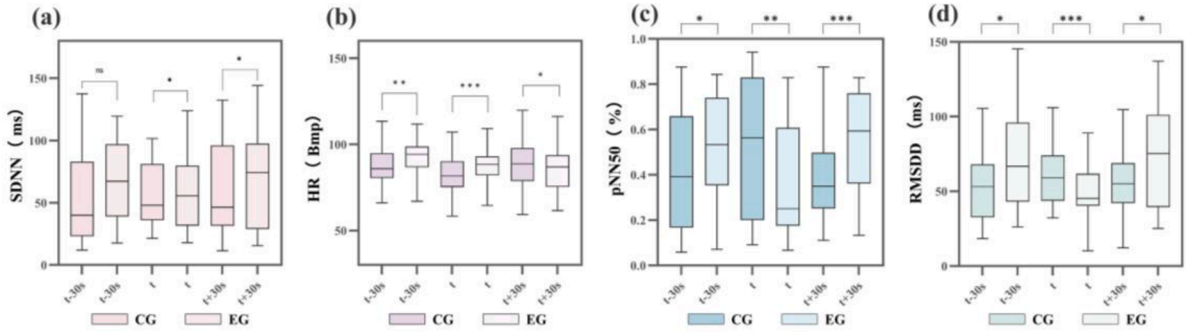


Fig. 13. Results of physiological data analysis. This figure presents boxplots of participants' physiological responses, including SDNN, heart rate (HR), PNN50, and RMSSD, measured at three time points: 30 s before fault occurrence (T–30 s, steady state), at fault occurrence (T), and 30 s after the fault (T + 30 s, recovery phase), for the control group (CG) and experimental group (EG).HR reflects overall physiological arousal, whereas SDNN, RMSSD, and PNN50 are heart rate variability (HRV) indices related to autonomic nervous system regulation and stress recovery. In each boxplot, the central line represents the median, the box indicates the interquartile range, and the whiskers show the data range. Statistical significance is indicated as ns (not significant), $p < 0.05$ (*), $p < 0.01$ (**), and $p < 0.0001$ (***).

and prioritizing critical alerts, noting that "*I'd rather have fewer but more important signals highlighted in the main field of view*" (P18).

Differences also emerged in decision-making preferences depending on operator experience and resilience levels. Experienced operators tended to prefer maintaining manual control supported by system-generated suggestions. As one participant explained, "*I feel more resilient when the system gives me options and I choose what to do, rather than it just making the decision for me*" (P05). In contrast, less experienced users preferred stronger system guidance: "*I'd rather the system just tells me what to do, because I might panic or hesitate if I have to decide myself*" (P26).

Participants particularly highlighted the advantages of the resilience-oriented interface used in the experimental group. Many reported that step-by-step suggestions and choice-based prompts helped them regain composure and confidence after encountering system faults. For example, one participant noted, "*The system gives me guidance that helps me calm down and figure out what to do next without freezing*" (P09), while another explained, "*I can see multiple solutions and decide based on the situation, which helps me manage stress and remain effective*" (P03). Others similarly reported that the interface allowed them to maintain decision authority while receiving reliable guidance, which supported rapid psychological recovery under stress.

Participants with prior SAR simulation experience also noted that the MR interface helped them combine experience-based judgment with structured guidance, further improving decision confidence. One participant commented, "*In the simulation, I could sense the environment better, which made my reactions more natural and confident*" (P04).

For future system optimization, participants suggested concentrating critical prompts within the main visual field, integrating countdown timers, and providing multi-strategy recommendations to support decision-making under pressure. Several participants also recommended adaptive assistance levels based on operator experience or stress level. As one participant stated, "*If the system can adjust its assistance based on stress level or experience, that would really help in real missions*" (P05). In addition, many highlighted the potential of MR systems for training purposes, noting that immersive simulations help build mental models and improve preparedness for real emergencies.

4.8.5. Threats to validity and mitigation strategies

To ensure the robustness of our findings, we considered potential threats to validity and implemented the following mitigation strategies throughout our experimental design and analysis.

Internal Validity: To minimize confounding factors, we employed a within-subjects design where each participant experienced both the control and experimental interfaces. The order of conditions was balanced across participants to counter potential learning or fatigue effects. All experiments were conducted in a controlled laboratory setting with standardized equipment and procedures, reducing environmental variability.

External Validity: Our participant pool (N = 35) included individuals with varying levels of UAV operation experience, as detailed in the Appendix A**-1**, which enhances the generalizability of our results across novice and experienced users. However, we acknowledge that the laboratory-based task, while designed to simulate risk-conflict scenarios, may not fully capture all complexities of real-world operations. Future work should validate these findings in field settings.

Construct Validity: Construct validity was strengthened through multi-method triangulation. Cognitive workload and situation awareness were assessed using established scales (NASA-TLX and SART). Psychological resilience was measured using an adapted CD-RISC scale (Appendix B), whose internal structure was examined through correlation analysis (Section 4.8.2). Subjective assessments were further complemented by objective physiological data and performance metrics to provide a more comprehensive evaluation of operator states.

Statistical Conclusion Validity: All statistical analyses were preceded by assumption checks, including normality tests using the Shapiro–Wilk test. Paired-sample *t*-tests were applied for within-subject comparisons, consistent with the experimental design. Outliers exceeding three standard deviations were removed using a consistent rule across datasets. Visual inspections, including boxplots and density checks, were also conducted to ensure robustness. Interpretations therefore emphasize consistent patterns across subjective, behavioral, and physiological measures rather than isolated significant results.

These strategies increase confidence that the observed differences between the control and experimental conditions are attributable to the resilience-enhancing interface design.

5. Discussion

This section provides a comprehensive summary of the study and discusses the findings based on experimental data analysis to validate the proposed research hypotheses. Leveraging the constructed design framework and prototype, a swarm human-machine collaboration simulator for search and rescue operations was developed. The study aimed to examine whether resilient interaction interfaces could effectively reduce operators' cognitive load and enhance their situation awareness and psychological resilience in risk-conflict scenarios. The effectiveness of the proposed interface design was assessed through an integrated analysis of subjective and objective measures.

5.1. Situation awareness and cognitive load

Previous studies have indicated that limited situational awareness (SA) in outdoor rescue scenarios can negatively affect the efficiency and success of rescue operations (Fernandez Rojas et al., 2019). To address

this challenge, this study proposed a resilience-oriented interaction framework and developed an MR-based HSI system designed to support operators' situational awareness while reducing cognitive load during risk-conflict scenarios.

The findings suggest that the resilience-oriented interface improves operators' understanding of situational uncertainty and variability, enabling them to manage complex tasks more effectively. Prior research has also explored XR-based human–machine interaction systems for enhancing situational awareness in UAV swarm search-and-rescue operations (Nalamothu et al., 2024). Building on these studies, the present work further emphasizes risk-conflict events and operators' psychological resilience. Earlier research has also proposed collaborative UAV swarm approaches based on situational awareness consensus and incorporated SA into interface design frameworks (Wattearachchi et al., 2025b). Extending these approaches, this study integrates two additional resilience dimensions—adaptation and recovery—into the traditional three-layer SA framework, providing a broader perspective for future situational awareness research.

Previous studies rarely examined how SA can be maintained when operators experience fluctuating stress or cognitive instability. The results of this study indicate that resilience-oriented interface structures can help stabilize situational awareness under such conditions by supporting clearer interpretation of task states and system feedback.

Cognitive load was evaluated using the NASA-TLX scale and physiological indicators. Overall, the resilience-oriented interface appeared to reduce perceived effort and negative emotional responses during risk-conflict tasks, while supporting operational performance. Physiological indicators such as SDNN, pNN50, and RMSSD further suggest improved physiological recovery during risk events and post-event stages. These observations are consistent with prior research that uses physiological measures to evaluate cognitive workload and stress recovery (Kothig, 2021; Wang et al., 2024; Pereira et al., 2025).

User interviews provided additional insight into the remaining limitations of the interface design. Previous research has suggested that interface transparency and task-load distribution can significantly influence user stress levels (Fletcher and Sarkar, 2013) . In the current system, several design factors may have increased psychological and physical demands. Participants reported that the positioning of the 3D sandbox and interface elements required frequent movement within the VR environment, increasing physical effort. Some interaction operations also showed delayed responsiveness, while inconsistent information prompts occasionally complicated task execution. These findings suggest that future designs should improve interface layout consistency, interaction responsiveness, and information presentation to further reduce operator workload.

5.2. Resilient design

When facing unexpected events, humans demonstrate adaptive and recovery capabilities that support stress resilience (Feder et al., 2009). In complex and uncertain environments such as outdoor rescue operations, the resilience of both operators and interaction systems plays a critical role. Existing research has largely focused on improving the resilience of UAV swarms or intelligent systems in rescue missions (Phadke and Medrano, 2024; Yu et al., 2024), while comparatively little attention has been given to operators' psychological resilience.

The findings of this study suggest that interaction design itself can contribute to strengthening operator resilience. By structuring information related to risk-conflict events and providing adaptive guidance, the interface supports operators in maintaining cognitive stability and managing stress during mission execution. This perspective complements system-level resilience research by emphasizing the human-centered dimension of resilience in human–machine collaboration.

To further examine psychological resilience, this study combined CD-RISC and ER89 scales with physiological indicators and response-time analysis. The integrated results indicate that resilience-oriented interfaces can support psychological stability and stress regulation during risk-conflict events. Physiological measures associated with autonomic nervous system activity suggest improved recovery and regulation processes, indicating that interaction support may facilitate cognitive and emotional stabilization under pressure.

Response-time analysis further suggests that resilience-oriented interfaces may support more efficient decision-making during later task stages. However, improvements were less evident during early detection and task-switching phases. One possible explanation is that additional resilience-related information may increase cognitive processing demands in early stages, highlighting the need to balance informational support with interaction efficiency in future designs.

Several resilience models have been proposed in previous studies, including the prediction–diagnosis–experimentation reasoning model (Hollnagel et al., 2006) and the prediction–detection–recovery resilience cycle (Amalberti, 2006). Building on these theoretical foundations, this study proposes a five-level human–machine resilience model consisting of Perception–Understanding–Prediction–Adaptation–Recovery. Correlation analysis indicates that adaptation and recovery dimensions are most strongly associated with overall resilience. This suggests that interface mechanisms supporting these two stages—such as adaptive guidance and recovery feedback—may play a particularly important role in strengthening operator resilience. Compared with earlier frameworks emphasizing system robustness, this model integrates human cognitive recovery into the situational awareness process and offers a more human-centered perspective on resilience design.

Overall, the findings indicate that resilience-oriented interface structures can support operators in managing dynamic and high-risk scenarios more effectively. These insights offer theoretical and practical implications for the design of immersive interaction systems in UAV rescue operations.

5.3. MR HSI interface design

Previous studies integrating VAM technologies with UAV search-and-rescue systems have improved human–machine collaboration and enhanced command-center situational awareness (Salunkhe et al., 2025; Nalamothu et al., 2024). Consistent with these studies, participants in our experiment reported improved situational understanding and emotional stability when using the resilience-oriented interface. However, unlike prior work that mainly emphasized visual augmentation and control support, this study highlights how resilience-driven interaction structures contribute to cognitive stability and risk recovery in high-pressure tasks.

Due to technical constraints, the UAV rescue simulation system in this study was developed within a VR environment. The VR platform served as an experimental testbed for validating the proposed design and collecting user feedback on immersive interaction. Based on this feedback, the study further explores the feasibility of extending resilience-oriented interface strategies to MR environments.

VR and MR environments differ in both interaction modes and environmental representations. VR creates a fully virtual immersive scene, whereas MR overlays digital information onto the real world and is influenced by factors such as lighting, communication latency, and weather conditions. Moreover, the complexity of real rescue operations makes it difficult for VR simulations to completely replicate real-world environmental conditions.

Most participants reported that the resilience-oriented interface improved their understanding of UAV malfunction scenarios and mission status, helping them remain stable under high-pressure tasks and complete operations successfully. The immersive sandbox interaction also improved engagement and spatial awareness, enabling operators to better understand the location of trapped individuals and other spatial task elements.

Participants also noted the advantages of VR simulation systems for operator training in wilderness SAR scenarios. Previous studies have

demonstrated that virtual simulators can effectively support professional training by balancing interface usability, physical realism, and environmental fidelity (Pappas et al., 2025) Mixed-reality games have also been used to explore search-and-rescue scenarios (Khalaf et al., 2018), while MR-based indoor disaster simulations have supported rescue system design (Chae et al., 2023). These systems allow operators to familiarize themselves with task environments and develop spatial mental models before engaging in real operations.

The present findings extend these studies by showing that resilience-oriented interface structures may enhance not only spatial awareness but also psychological readiness and stress regulation. Consequently, VR and MR systems hold considerable potential for UAV rescue mission training and operational support, with future research expanding toward real-world rescue environments.

From a design perspective, MR interface systems should prioritize operators' situational awareness by integrating multi-source data from UAV sensors and providing alerts based on task urgency and risk levels. Operational guidance and contextual prompts within MR environments can support the construction of clear mental models and improve cognitive preparedness. During the adaptation stage, the system should provide structured interpretations of risk events and deliver step-by-step guidance that aligns with operators' cognitive needs. During the recovery stage, timely positive feedback can support physiological and psychological recovery, thereby strengthening the resilience of the overall mission workflow.

5.4. Limitations

The limitations of this study are mainly as follows:

First, the sample size is relatively small (a total of 35 participants), and the participants' demographic characteristics are relatively homogeneous (e.g., age, gender, UAV experience), which somewhat limits the generalizability of the research findings. Second, the physiological data collection equipment is sensitive to wear and signal transmission, and due to participant movement during the experiment, signal instability occurred. As a result, only 27 valid samples were retained for analysis, potentially affecting the representativeness and stability of the data. Third, although the VR simulation environment aimed to closely replicate the wilderness SAR scenario, it was still challenging to fully replicate the complex environmental variables present in real-life situations, which may have impacted the external validity of the study. Additionally, this study only used UAV malfunctions as a risk-conflict event, resulting in a relatively narrow contextual setup. Future studies should expand to a broader range of risk types in order to more comprehensively validate the applicability of resilience design strategies.

6. Conclusion

This study investigates risk-conflict scenarios in wilderness SAR missions, focusing on operators' situational awareness (SA) and psychological resilience in human–machine collaboration with UAV swarms. It systematically identifies sources of risk conflicts, constructs a cognitive model for HSI, and proposes a resilience design framework based on SA theory to guide interface development. The framework aims to enhance operators' immersive and interpretable interaction support in complex and uncertain rescue tasks.

Combining experimental data with subjective perceptions and objective physiological measures, the study validates the effectiveness of the proposed system in reducing cognitive load, enhancing situational understanding, and supporting psychological recovery. Results show that resilient HSI interface design improves operators' SA and promotes physiological stability and stress recovery, highlighting its effectiveness in complex rescue operations.

In terms of theoretical contributions, this study advances current knowledge by extending situational awareness theory into the domain of human–AI collaboration under risk, introducing a resilience-centered perspective that complements traditional cognition-driven interaction models. Building on classical SA frameworks, the research proposes a five-level human–machine resilience cognitive model (Perception–Comprehension–Prediction–Adaptation–Recovery), emphasizing how operators' adaptation and recovery abilities can be actively supported through interface design. The empirical evidence provided bridges a gap in existing literature, which traditionally focuses on system-level resilience, by demonstrating that psychological and cognitive resilience at the operator level can be enhanced through interaction design. This contributes a new theoretical lens for interpreting stability, adaptability, and recovery processes in human–machine teams.

Practically, our study offers a set of actionable design principles for developing MR-based HSI systems suited to complex and high-risk operational environments. The resilience-oriented design framework provides guidance on how to structure information presentation, risk interpretation, adaptive assistance, and recovery feedback within immersive systems. These insights are not limited to UAV rescue domains but can also inform the design of adaptive and stress-aware interfaces in other high-risk fields such as autonomous driving, disaster response, aerospace operations, and remote robotic control. Additionally, the VR-based simulation and evaluation process demonstrates a feasible pathway for iterative interface development and training in scenarios where field testing is difficult or dangerous.

Overall, this study not only provides a new perspective on interaction design in HSI systems but also lays a theoretical foundation and practical reference for the future application of resilient interaction design in other high-risk scenarios.

CRediT authorship contribution statement

Fang You: Supervision. **Yuqing Jiang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation. **Siqi Pan:** Validation. **Yuchen Wang:** Software. **Qianwen Fu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Semi-structured interview questions and representative responses

This semi-structured interview is designed to last approximately 15 min. Due to the limited interview time and the need to accommodate participants' varying task experiences, not all questions will be asked in every session. Sections 1–3 contain mandatory questions that will be asked to all participants in order to ensure consistency in core data collection. Sections 4–8 include optional questions, which may be selectively asked depending on the flow of the conversation, the participant's responses, and the available time. The interviewer may adjust the order or depth of the questions as

needed while ensuring that all mandatory questions are covered.

A-1. Semi-structured interview

Theme	Sample Questions
Section 1. **Introduction and Warm-Up**	**Q1.** Before we begin, may I ask whether you have any prior experience with drones, automated systems, or similar human–machine interfaces?
	Q2. During today's task, did you feel clear about your role and responsibilities in the human–machine collaboration process
Section 2. **Task Recall**	**Q3.** Can you describe the two key events you encountered during the simulation? (a) Identifying a suspected victim (b) Responding to a UAV malfunction
	Q4. During these events, what specific pieces of information did you focus on the most?
Section 3. **Perceived Differences Between the Two groups**	**Q5.** What differences did you notice between the first condition and the second condition in terms of information presentation?
	Q6. Did you observe differences in how explanations, options, or confirmations were provided between the two groups?
	Q7. In the UAV malfunction scenario, how did you perceive the difference between: What were your impressions of this contrast?
Section 4. **Human–Machine Decision Authority Preferences**	**Q8.** In a risk-conflict situation like UAV malfunction, which decision-making arrangement did you prefer? Why?
	Q9. Did you feel that the decision-support style in Condition 2 (explanation + options + confirmation) increased your confidence or increased your cognitive load? How so?
	Q10. Did you feel any form of information overload in either condition? If yes, can you describe when and why?
Section 5. **Psychological Resilience and Emotional Reactions**	**Q11.** How did you feel emotionally when you first encountered the UAV malfunction (e.g., stress, panic, confusion)?
	Q12. Between the two conditions, which one helped you recover from stress or regain composure more effectively? Why?
	Q13. Did explanations or transparency from the system help reduce your uncertainty or anxiety? Please explain.
Section 6. **Usability and Interface Experience**	**Q14.** Did you experience any difficulties or usability issues while interacting with the interface? (e.g., unreadable text, slow response, too many pop-up windows, confusing layout)
	Q15. Did you find the interface layout and explanation structure in both conditions clear enough? If not, how should they be improved?
Section 7. Overall Preferences and System Expectations	**Q16.** If this system were deployed in a real search-and-rescue mission, which interaction style would you prefer and why?
Section 8. Closing	**Q17.**What improvements would you suggest for enhancing psychological resilience in future human–machine collaboration systems?

A-2. Thematic analysis results of the interviews

This table summarizes the typical feedback provided by participants while using the mixed reality (MR) UAV operation system. Based on the participants' responses, we conducted a thematic analysis of the interview data, which identified six main themes: immersion and intuitive operation, information and visual overload, decision-making preferences, psychological resilience, search and rescue (SAR) simulation experience, and system optimization suggestions.

Topic	Typical Quotes	Participant Type	Participant Number
Immersion and Intuitive Operation	"Compared to just clicking on a 2D screen, the 3D interface makes it easier to see what's happening around the UAV, and switching perspectives really helps me understand the situation faster."	UAV General Experience	P07
	"The MR view makes coordination easier because I can immediately understand where each UAV is relative to the valley area."	SAR Experience	P03
	"Switching between global and local perspectives helps me better understand the environmental conditions and the rescue targets."	UAV General Experience	P12
	"In 3D, I can instantly see if the UAV is near a hazard; in 2D, I might miss it."	UAV General Experience	P15
Information Overload and Visual Overload	"There's just too much information popping up in this risk scenario… I don't have the mental capacity to read all the text while managing the UAV."	UAV General Experience	P14
	"I'd rather have fewer but more important signals highlighted in the main field of view."	UAV General Experience	P18
	"Sometimes the prompts come too fast; I can't keep up and it makes me nervous."	UAV General Experience	P22
	"If all the warnings are in one place, I can react faster instead of scanning the screen."	SAR Experience	P04
Decision-Making Preference: Human-Assisted	"I feel more resilient when the system gives me options and I choose what to do, rather than it just making the decision for me."	UAV General Experience	P05
	"I trust the system to analyze, but I still want the final call—especially in emergencies."	SAR Experience	P03
	"Being able to decide which solution to implement makes me feel in control and reduces panic."	UAV General Experience	P10
	"I prefer when the system provides guidance but lets me execute the final step."	UAV General Experience	P16
Decision-Making Preference: System-Directed	"I'd rather the system just tells me what to do, because I might panic or hesitate if I have to decide myself."	UAV General Experience	P26
	"If the system makes the decision for me, I feel less anxious because I don't have to worry about making a wrong call."	UAV General Experience	P21
	"I feel safer when the machine takes responsibility in high-pressure scenarios."	UAV General Experience	P08
	"Sometimes I just want the system to act immediately without me thinking too much."	UAV General Experience	P11

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Topic	Typical Quotes	Participant Type	Participant Number
Psychological Resilience and Recovery	"The MR system gives me guidance that helps me calm down and figure out what to do next without freezing."	UAV General Experience	P09
	"I can see multiple solutions and decide based on the situation, which helps me manage stress and remain effective."	SAR Experience	P03
	"Having suggestions to follow step by step reduces my anxiety."	UAV General Experience	P17
	"Even in tense moments, the system's prompts help me recover mentally faster."	UAV General Experience	P20
Advantages of SAR Simulation Experience	"In the simulation, I could sense the environment better, which made my reactions more natural and confident."	SAR Experience	P04
	"Having multiple solutions in front of me and choosing based on experience makes me feel more in control under pressure."	SAR Experience	P05
	"The MR interface lets me apply prior knowledge effectively in rescue scenarios."	SAR Experience	P01
	"Compared to traditional controls, MR helps me manage several UAVs without losing track."	SAR Experience	P02
System Optimization Suggestions	"If the system can adjust its assistance based on stress level or experience, that would really help in real missions."	SAR Experience	P05
	"Practicing with the MR interface helps me understand the situation better, and I feel more confident facing real emergencies"	UAV General Experience	P09
	"Critical alerts should appear directly in the main view so I can react faster."	UAV General Experience	P14

Appendix B. Evaluation scales

Appendix B provides the subjective evaluation scales used in Section 4.7 of this study. These scales were employed to assess participants' situational awareness, cognitive load, and psychological resilience. The CD-RISC Scale and ER89 Scale were appropriately adapted to align with the experimental context and task characteristics of this study. Specifically, item wordings were modified to better reflect the environmental demands and task scenarios encountered in the UAV-based search and rescue setting.

B-1. Situational awareness (SART)

This section measured participants' situational awareness during the UAV rescue operation.
All items were measured on a seven-point Likert scale (1 = very low, 7 = very high).

Domains	Construct	Item
Attentional demand	Instability of situation	How unstable was the task environment during the SAR (search and rescue) mission? Was it relatively stable and predictable (low) or highly unstable with frequent sudden changes (high)?
	Variability of situation	How complex was the SAR mission? Was it simple and straightforward (low) or involved many interrelated factors (high)?
	Complexity of situation	How much variation occurred in the task environment during the SAR mission? Were there few changing variables (low) or many factors changing constantly (high)?
Attentional supply	Arousal	How alert were you when performing the SAR task? Were you less alert to unexpected events (low) or highly vigilant and ready to respond (high)?
	Spare mental capacity	How focused was your attention during the SAR mission? Were you mainly concentrating on a single aspect (low) or simultaneously attending to multiple aspects (high)?
	Concentration	Was your attention divided during the rescue operation? Were you able to focus on the main task (low) or needed to attend to multiple task elements (high)?
	Division of attention	How much spare mental capacity did you have while handling the SAR task? Did you feel cognitively overloaded (low) or have sufficient mental resources to manage multiple task demands (high)?
Understanding	Information quantity	How much information did you acquire during the mission? Did you receive little information (low) or a large amount of relevant information (high)?
	Information quality	How would you rate the quality of the information received? Was it of low quality (low) or highly relevant and accurate (high)?
	Familiarity	How familiar were you with the SAR task? Was it your first experience (low) or did you have extensive prior experience (high)?

B-2. Cognitive workload (NASA task load index)

(7-point Likert scale: 1 = very low, 7 = very high)
This section evaluated participants' perceived cognitive workload during the UAV rescue task.

Domains	Item
Mental Demand	How much mental and perceptual activity was required (e.g., thinking, deciding, remembering, looking, searching)? Was the task easy or demanding, simple or complex, lenient or precise?
Physical Demand	How much physical activity was required (e.g., controlling, pushing, turning, adjusting)? Was the task easy or demanding, slow or rapid, relaxed or strenuous?

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Domains	Item
Temporal Demand	How much time pressure did you feel due to the pace of the rescue operation? Was it slow and relaxed, or rapid and hectic?
Performance	How successful were you in accomplishing the goals set by the experiment or yourself? How satisfied were you with your performance?
Effort	How hard did you have to work (mentally and physically) to accomplish your level of performance?
Frustration Level	How insecure, discouraged, irritated, or stressed versus secure, gratified, content, or relaxed did you feel during the task?

B-3. System usability scale

(5-point Likert scale: 1 = strongly disagree, 5 = strongly agree)
This section assessed the perceived usability of the 3D-VR UAV rescue system compared with a traditional 2D interface.

Number	Item
1	If I were a UAV operator in a disaster rescue mission, I would prefer using the 3D-VR rescue system rather than a 2D interface.
2	I find the 3D-VR UAV rescue system overly complex.
3	I find this system easy to use during rescue operations.
4	I think I would need assistance from others to use this system during the mission.
5	I find the system's functions well integrated during operation.
6	I find inconsistencies in the system's visual presentation and semantic expression.
7	I believe that most operators like me could learn to use this system quickly during rescue missions.
8	I think I would need to learn many additional skills or knowledge before using the system effectively.
9	The system feels smooth and responsive during operation.
10	The system's prompts and feedback effectively guide me to handle risks and problems encountered during the rescue mission.

B-4. Resilience (CD-RISC)

(5-point Likert scale: 1 = strongly disagree, 5 = strongly agree)
This section measured participants' psychological resilience during UAV rescue missions.

Number	Item
1	I can quickly adapt to changes in the environment and mission requirements.
2	I establish strong and reliable collaboration with team members.
3	I believe our mission has a deeper sense of purpose and support.
4	I am confident in my abilities to handle various challenges during the rescue.
5	Past successes give me confidence in achieving future success.
6	I remain optimistic even under stress, such as when UAV failures occur.
7	I believe moderate pressure can enhance my performance.
8	I can quickly adjust and continue working after a UAV failure.
9	I see every equipment failure as a chance to learn and improve.
10	I devote myself fully to overcoming challenges in rescue missions without giving up.
11	I firmly believe I can achieve rescue goals despite many obstacles.
12	I stay persistent and search for breakthroughs when rescue operations encounter difficulties.
13	I believe I can find solutions even when resources are limited.
14	I remain focused and analytical under pressure.
15	I am willing to take leadership roles in solving team problems.
16	I learn from failures and prepare for the next attempt.
17	I consider myself resilient when facing technical difficulties such as UAV failures.
18	I take responsibility for making difficult but beneficial decisions for the mission.
19	I can manage negative emotions effectively during rescue operations.
20	I trust my intuition to make the right decisions at critical moments.
21	I have a strong sense of mission and purpose in UAV rescue tasks.
22	I believe I can maintain control and guide the mission even in uncertain situations.
23	I enjoy challenges during rescue operations as opportunities for growth.
24	I continuously work toward achieving rescue goals.
25	I feel proud of completing rescue tasks and saving lives.

B-5. Ego-resilience (ER89)

(5-point Likert scale: 1 = strongly disagree, 5 = strongly agree)
This section assessed participants' adaptability, emotional recovery, and positive coping during the rescue mission.

Number	Item
1	I can generously share information and resources during UAV rescue missions.
2	When a UAV fails, I can recover quickly from the initial shock.
3	I respond positively to new challenges and situations in the rescue process.
4	I usually leave a positive impression when communicating with team members.
5	I am willing to adopt new technologies and methods to improve rescue efficiency.
6	I maintain continuous energy and motivation throughout the mission.
7	I am open to trying different rescue strategies, even in familiar environments.
8	I stay highly curious about new information and solutions during the mission.
9	I generally trust the team members I collaborate with.
10	I carefully consider all potential risks and outcomes when planning rescue operations.
11	I enjoy experimenting with innovative rescue approaches and strategies.
12	My rescue missions provide continuous opportunities for learning and adapting to new challenges.
13	I believe I possess strong psychological resilience when facing risks such as UAV malfunctions.
14	When encountering setbacks, I can quickly adjust my emotions and move forward.

Data availability

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

References

Abbas, S., Jeong, H., 2024. Task difficulty impact on multitasking in mixed reality environments. Comput. Educ. X Real. 4, 100065.

Abioye, A.O., Hunt, W., et al., 2024. The effect of predictive formal modelling at runtime on performance in human-swarm interaction. In: Proceedings of the Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction.

Abioye, A.O., Hunt, W., et al., 2025. A user study evaluation of predictive formal modelling at runtime in human-swarm interaction. ACM Trans. Hum. Robot. Interact. 14 (4), 1–24.

Alqudsi, Y., Makaraci, M., 2025. UAV swarms: research, challenges, and future directions. J. Eng. Appl. Sci. 72 (1), 12.

Amalberti, R., 2006. Optimum system safety and optimum system resilience. In: Hollnagel, E., Woods, D., Leveson, N. (Eds.), Resilience Engineering. Ashgate, Aldershot.

Angelopoulos, A., Hale, A., et al., 2022. Drone brush: mixed reality drone path planning. In: Proceedings of the 17th ACM/IEEE International Conference on Human-Robot Interaction (HRI). IEEE.

Bangor, A., Kortum, P., et al., 2009. Determining what individual SUS scores mean: adding an adjective rating scale. J. Usability Stud. 4 (3), 114–123.

Bashyal, S., Venayagamoorthy, G.K., 2008. Human swarm interaction for radiation source search and localization. In: Proceedings of the IEEE Swarm Intelligence Symposium. IEEE.

Benabbou, F.B.E., et al., 2023 "A survey of research progress and emerging trends in human-swarm interactions." Available at SSRN 4424713.

Bhardwaj, R., Balasubramanian, V., 2019. Viability of cardiac parameters measured unobtrusively using capacitive coupled electrocardiography (cECG) to estimate driver performance. IEEe Sens. J. 19 (11), 4321–4330.

Billinghurst, M., Duenser, A., 2012. Augmented reality in the classroom. Computer 45 (7), 56–63. . (Long. Beach. Calif).

Bobko, P., Hirshfield, L., et al., 2023. Human-agent teaming and trust calibration: a theoretical framework, configurable testbed, empirical illustration, and implications for the development of adaptive systems. Theor. Issues Ergon. Sci. 24 (3), 310–334.

Bruce, J., Perron, J., et al., 2017. Ready—Aim—Fly! hands-free face-based HRI for 3D trajectory control of UAVs. In: Proceedings of the 14th Conference on Computer and Robot Vision (CRV). IEEE.

Catellani, M., Nironi, F., et al., 2024. A mixed reality interface for human-swarm interaction. European Robotics Forum. Springer.

Chae, Y., Lee, H., et al., 2023. Design of a mixed reality system for simulating indoor disaster rescue. Appl. Sci. 13 (7), 4418.

Clerigues, D., Wubben, J., et al., 2024. Enabling resilient UAV swarms through multi-hop wireless communications. EURASIP J. Wirel. Commun. Netw. 2024 (1), 39.

Connor, K.M., Davidson, J.R., 2003. Development of a new resilience scale: the Connor-Davidson resilience scale (CD-RISC). Depress. Anxiety 18 (2), 76–82.

Coovert, M.D., Lee, T., et al., 2014. Spatial augmented reality as a method for a mobile robot to communicate intended movement. Comput. Human. Behav. 34, 241–248.

Divband Soorati, M., Clark, J., et al., 2021. Designing a user-centered interaction interface for human–swarm teaming. Drones 5 (4), 131.

Endsley, M.R., 1988. Design and evaluation for situation awareness enhancement. In: Proceedings of the Human Factors Society Annual Meeting. Los Angeles, CA. Sage Publications Sage CA.

Endsley, M.R., 1995. Toward a theory of situation awareness in dynamic systems. Hum. Factors 37 (1), 32–64.

Endsley, M.R. (1999). "Situation awareness in aviation systems." Handbook of aviation human factors 11: 257–276.

Ewers, J., Anderson, D., et al., 2025. Deep reinforcement learning for time-critical wilderness search and rescue using drones. Front. Robot. AI 11, 1527095.

Farkas, D., Orosz, G., 2015. Ego-resiliency reloaded: a three-component model of general resiliency. PLoS One 10 (3), e0120883.

Feder, A., Nestler, E.J., et al., 2009. Psychobiology and molecular genetics of resilience. Nat. Rev. Neuroscience. 10 (6), 446–457.

Fernandez Rojas, R., Debie, E., et al., 2019. Encephalographic assessment of situation awareness in teleoperation of human-swarm teaming. In: Proceedings of the Neural Information Processing: 26th International Conference, ICONIP 2019, Sydney, NSW, Australia, December 12–15, 2019, Proceedings, 26. Springer.

Fletcher, D., Sarkar, M., 2013. Psychological resilience: a review and critique of definitions, concepts, and theory. Eur. Psychol. 18 (1), 12–23.

Fletcher, P., Luther, A., et al., 2022. Predicting visual differentiability for unmanned aerial vehicle gestures. IEEe Robot. Autom. Lett. 7 (4), 11799–11806.

Flood, A., Keegan, R.J., 2022. Cognitive resilience to psychological stress in military personnel. Front. Psychol. 13, 809003.

Fong, T., Thorpe, C., et al., 2003. Robot, asker of questions. Rob. Auton. Syst. 42 (3–4), 235–243.

Gao, Q., Yang, Y., et al., 2022. EEG-based emotion recognition with feature fusion networks. Int. J. Mach. Learn. Cybern. 13 (2), 421–429.

Gao, Y., Li, D., 2020. Unmanned aerial vehicle swarm distributed cooperation method based on situation awareness consensus and its information processing mechanism. Knowl. Based Syst. 188, 105034.

Gao, Y., Li, D.S., et al., 2018. UAV distributed swarm situation awareness model. J. Electr. Inf. Technol. 40 (6), 1271–1278.

Granadeno, P.A.A. and J. Cleland-Huang (2025). "Land-coverage aware path-planning for multi-UAV swarms in search and rescue scenarios." arXiv preprint arXiv: 2505.08060.

Gu, Y., Hunt, W., et al., 2023. Successful swarms: operator situational awareness with modelling and verification at runtime. In: Proceedings of the 32nd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN). IEEE.

Guo, Y., Pang, Y., et al., 2024. Trust-aware reflective control for fault-resilient dynamic task response in Human–Swarm cooperation. AI 5 (1), 446–464.

Han, W., Wang, Y., et al., 2024. A Framework for UAV Swarm Situation Awareness. Springer Nature Singapore, Singapore.

Hao, T., Zheng, X., et al., 2022. Linear and nonlinear analyses of heart rate variability signals under mental load. Biomed. Signal. Process. Control 77, 103758.

Hart, S.G., Staveland, L.E., 1988. Development of NASA-TLX (Task Load Index): results of empirical and theoretical research. In: Advances in Psychology, 52. Elsevier, pp. 139–183.

Healey, J.A., Picard, R.W., 2005. Detecting stress during real-world driving tasks using physiological sensors. IEEE Trans. Intell. Transp. Syst. 6 (2), 156–166.

Hoang, M.O. and K.A.R. Grøntved, et al. (2023). "Drone swarms to support search and rescue operations: opportunities and challenges." Cultural Robotics: Social Robots and Their Emergent Cultural Ecologies: 163–176.

Hocraffer, A., Nam, C.S., 2017. A meta-analysis of human-system interfaces in unmanned aerial vehicle (UAV) swarm management. Appl. Ergon. 58, 66–80.

Hollnagel, E., 2014. Resilience engineering and the built environment. Build. Res. Inf. 42 (2), 221–228.

Hollnagel, E., Woods, D.D., et al., 2006. Resilience engineering: Concepts and Precepts. Ashgate Publishing, Ltd.

Hou, Y., Xie, Q., et al., 2025. Cognitive load classification of mixed reality human computer interaction tasks based on multimodal sensor signals. Sci. Rep. 15 (1), 13732.

Irshad, L., Hulse, D., 2022. Resilience modeling in complex engineered systems with human-machine interactions. In: Proceedings of the International Design Engineering Technical Conferences and Computers and Information in Engineering Conference. American Society of Mechanical Engineers.

İşler, Y., Kuntalp, M., 2007. Combining classical HRV indices with wavelet entropy measures improves to performance in diagnosing congestive heart failure. Comput. Biol. Med. 37 (10), 1502–1510.

Johansson, B.J., Lundberg, J., 2025. No robot is an island-what properties should an autonomous system have in order to be resilient? Theor. Issues Ergon. Sci. 26 (2), 197–216.
Jones, S., Milner, E., et al., 2020. Distributed situational awareness in robot swarms. Adv. Intell. Syst. 2 (11), 2000110.
Kapellmann-Zafra, G., Salomons, N., et al., 2016. Human-robot swarm interaction with limited situational awareness. In: Proceedings of the Swarm Intelligence: 10th International Conference, ANTS 2016, Brussels, Belgium, September 7-9, 2016, Proceedings, 10. Springer.
Khalaf, A.S., Pianpak, P., et al., 2018. An architecture for simulating drones in mixed reality games to explore future search and rescue scenarios. In: Proceedings of the ISCRAM 2018 Conference Proceedings–15th International Conference on ….
Kolling, A., et al., 2013. Human swarm interaction: an experimental study of two types of interaction with foraging swarms. J. Hum. Robot Interact. 2 (2), 103–129.
Kolling, A., Walker, P., et al., 2015. Human interaction with robot swarms: a survey. IEEe Trans. Hum. Mach. Syst. 46 (1), 9–26.
Kothig, A., 2021. Accessible Integration of Physiological Adaptation in Human-Robot Interaction. University of Waterloo.
Langerwisch, M., Wittmann, T., et al., 2013. Heterogeneous teams of unmanned ground and aerial robots for reconnaissance and surveillance-a field experiment. In: Proceedings of the IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR). IEEE.
Lewis, M., 2013. Human interaction with multiple remote robots. Rev. Hum. Factors Ergonom. 9 (1), 131–174.
Liu, C., Shen, S., 2020. An augmented reality interaction interface for autonomous drone. In: Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE.
Liu, Y., Nejat, G., 2016. Multirobot cooperative learning for semiautonomous control in urban search and rescue applications. J. Field. Robot. 33 (4), 512–536.
Lucchesi, L., Endler, M., et al., 2024. Dynamic effects of communication delay, failure rates, and speed on UAV swarm formation. In: Proceedings of the Simpósio Brasileiro de Redes de Computadores e Sistemas Distribuídos (SBRC), SBC.
Ludwig, D., Walker, B., et al., 1997. Sustainability, stability, and resilience. Conserv. Ecol. 1 (1).
Lyons, J.B., Capiola, A., et al., 2025. Examining the human-centred challenges of human–swarm interaction. Philos. Trans. A 383 (2289), 20240140.
Ma, X., Liu, P., et al., 2021. EEG emotion recognition based on optimal feature selection. In: Proceedings of the Journal of Physics: Conference Series. IOP Publishing.
Macchini, M., Schiano, F., et al., 2019. Personalized telerobotics by fast machine learning of body-machine interfaces. IEEe Robot. Autom. Lett. 5 (1), 179–186.
Meshkati, N., Loewenthal, A., 1988. An eclectic and critical review of four primary mental workload assessment methods: a guide for developing a comprehensive model. Adv. Psychol. 52, 251–267.
Morando, L., Zhou, X., et al., 2025. Human-drone collaboration via mixed-reality for efficient navigation and interaction in constrained environments: a comprehensive user case study. Auton. Robots 49 (4), 40.
Nagi, J., Giusti, A., et al., 2014. Human control of UAVs using face pose estimates and hand gestures. In: Proceedings of the ACM/IEEE International Conference on Human-Robot Interaction.
Nalamothu, R., Sontha, P., et al., 2024. Leveraging augmented reality for improved situational awareness during UAV-driven search and rescue missions. In: Proceedings of the IEEE International Symposium on Safety Security Rescue Robotics (SSRR). IEEE.
Navas-Medrano, S., Soler-Dominguez, J.L., et al., 2024. Mixed reality for a collective and adaptive mental health metaverse. Front. Psychiatry 14, 1272783.
Pajares, G., 2015. Overview and current status of remote sensing applications based on unmanned aerial vehicles (UAVs). Photogramm. Eng. Remote Sens. 81 (4), 281–330.
Pappas, G., et al., 2025. Generic drone simulator: design, development and user testing in a virtual search and rescue mission. IEEe Comput. Graph. Appl. 46 (2), 98–111.
Parker, L.E., 2007. Distributed Intelligence: overview of the field and its application in multi-robot systems. In: Proceedings of the AAAI Fall Symposium: Regarding the Intelligence in Distributed Intelligent Systems.
Pereira, E., Sigcha, L., et al., 2025. Capturing mental workload through physiological sensors in human–robot collaboration: a systematic literature review. Appl. Sci. 15 (6), 3317.
Phadke, A., Medrano, F.A., 2022. Towards resilient UAV swarms—A breakdown of resiliency requirements in UAV swarms. Drones 6 (11), 340.
Phadke, A., Medrano, F.A., 2024. Increasing operational resiliency of uav swarms: an agent-focused search and rescue framework. Aerosp. Res. Commun. 1, 12420.
Reardon, C., Lee, K., et al., 2019. Augmented reality for human-robot teaming in field environments. In: Proceedings of the Virtual, augmented and mixed reality. Applications and case studies: 11th International Conference, VAMR 2019, Held as Part of the 21st HCI International Conference, HCII 2019, Orlando, FL, USA, July 26–31, 2019, Proceedings, 21. Springer.
Rosen, E., Whitney, D., et al., 2020. Communicating robot arm motion intent through mixed reality head-mounted displays. In: Proceedings of the Robotics Research: The 18th International Symposium ISRR. Springer.
Salas, E., Prince, C., et al., 1995. Situation awareness in team performance: implications for measurement and training. Hum. Factors 37 (1), 123–136.
Salunkhe, S.A. and P. Nedunghat, et al. (2025). "Intuitive human-drone collaborative navigation in unknown environments through mixed reality." arXiv preprint arXiv: 2504.01350.
Samarasinghe, D., 2023. Counterfactual learning in enhancing resilience in autonomous agent systems. Front. Artif. Intell. 6, 1212336.
Sato, N., Matsuno, F., et al., 2004. Cooperative task execution by a multiple robot team and its operators in search and rescue operations. In: Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)(IEEE Cat. No. 04CH37566). IEEE.
Sende, M., Raffelsberger, C., et al., 2024. Bridging the reality gap in drone swarm development through mixed reality. Auton. Robots 48 (7), 19.
Sheridan, T.B. and W.L. Verplank (1978). "Human and Computer Control of Undersea Teleoperators.".
Siean, A., Gradinaru, B., et al., 2023. Opportunities and challenges in human-swarm interaction: systematic review and research implications. Int. J. Adv. Comput. Sci. Appl. 14 (4), 896–902.
Singh, R.R., Conjeti, S., et al., 2013. A comparative evaluation of neural network classifiers for stress level analysis of automotive drivers using physiological signals. Biomed. Signal. Process. Control 8 (6), 740–754.
Stanton, N.A., Salmon, P.M., et al., 2017. State-of-science: situation awareness in individuals, teams and systems. Ergonomics 60 (4), 449–466.
Steen, R., Aven, T., 2011. A risk perspective suitable for resilience engineering. Saf. Sci. 49 (2), 292–297.
Suzuki, Y., Wild, F., et al., 2024. Measuring cognitive load in augmented reality with physiological methods: a systematic review. J. Comput. Assist. Learn. 40 (2), 375–393.
Sweller, J., 1988. Cognitive load during problem solving: effects on learning. Cogn. Sci. 12 (2), 257–285.
Taub, G., Elmalech, A., et al., 2024. Trust and attitude toward information presented using augmented reality and other technological means. Heliyon 10 (4).
Taylor, R.M., 2017. Situational awareness rating technique (SART): the development of a tool for aircrew systems design. Situational Awareness. Routledge, pp. 111–128.
Tyler, W.J., Adavikottu, A., et al., 2025. Neurotechnology for enhancing human operation of robotic and semi-autonomous systems. Front. Robot. AI 12, 1491494.
Van Merriënboer, J.J., Sweller, J., 2010. Cognitive load theory in health professional education: design principles and strategies. Med. Educ. 44 (1), 85–93.
Walker, M., Hedayati, H., et al., 2018. Communicating robot motion intent with augmented reality. In: Proceedings of the ACM/IEEE International Conference on Human-Robot Interaction.
Walker, M., Chen, Z., et al., 2021. A mixed reality supervision and telepresence interface for outdoor field robotics. In: Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE.
Wang, P., Houghton, R., et al., 2024. Detecting and predicting pilot mental workload using heart rate variability: a systematic review. Sensors 24 (12), 3723.
Wattearachchi, W.D. and E. Lakshika, et al. (2025a). "A study on human-swarm interaction: a framework for assessing situation awareness and task performance." arXiv preprint arXiv:2503.14810.
Wattearachchi, W.D. and E. Lakshika, et al. (2025b). "Designing effective human-swarm interaction interfaces: insights from a user study on task performance." arXiv preprint arXiv:2504.02250.
Westrum, R., 2017. A typology of resilience situations. Resilience Engineering. CRC Press, pp. 55–65.
Xu, Q., Nwe, T.L., et al., 2014. Cluster-based analysis for personalized stress evaluation using physiological signals. IEEe J. Biomed. Health Inform. 19 (1), 275–281.
Yu, Y., Lakemond, N., et al., 2024. Resilience in emerging complex intelligent systems: a case study of search and rescue. J. Contingencies Crisis Manag. 32 (4), e12626.
Zhang, X., Liu, J., 2023. Research on UAV swarm network modeling and resilience assessment methods. Sensors 24 (1), 11.
Zhao, L., Chen, B., et al., 2025. Research on swarm control based on complementary collaboration of unmanned aerial vehicle swarms under complex conditions. Drones 9 (2), 119.
Zheng, C., Jarecki, A., et al., 2023. Integrated system architecture with mixed-reality user interface for virtual-physical hybrid swarm simulations. Sci. Rep. 13 (1), 14761.
Zhou, J., Zhu, H., Kim, M., Cummings, M.L., 2019. The impact of different levels of autonomy and training on operators' drone control strategies. ACM Trans. Hum.-Robot Interact. 8 (4), 1–15.
Zieba, S., Polet, P., et al., 2010. Principles of adjustable autonomy: a framework for resilient human–machine cooperation. Cogn. Technol. Work 12 (3), 193–203.